

From Rheology to Scaffold

*Engineering Principles and Operating Guide for Direct Ink Writing of CMC /
Sunflower-Oil Nanoemulsion Bioinks*

*Theory, dimensionless analysis, three worked-example inks, failure morphology, nanoemulsion
mechanics, biological considerations, lab protocols, and troubleshooting*

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Who this document is for and how to use it.

This is a self-contained engineering guide to printing CMC and CMC + sunflower-oil nanoemulsion bioinks on the piston-driven DIW gel printer. It is written for a new lab member or intern with an undergraduate engineering background (basic calculus, basic fluid mechanics) and no prior rheology experience. It is also a reference for experienced practitioners seeking a rigorous foundation for material selection, process parameter estimation, failure diagnosis, and regulatory justification.

The document is in four parts:

- **Part I – Theory** (Ch. 1–7): step-by-step engineering framework connecting rheological measurements to printing parameters. Includes failure morphology and dimensionless analysis.
- **Part II – Inks and Instruments** (Ch. 8–11): ink characterisation, nanoemulsion mechanics, rheological testing protocols, and extended material classes.
- **Part III – Protocols** (Ch. 12–17): MATLAB and Python workflows, calibration-cube sequence, and troubleshooting decision tree.
- **Part IV – Extended Topics** (Ch. 18–19): biological and regulatory considerations; statistical optimisation methods.

Navigation guide. If you only need to print and trust the upstream calibration, skip to Part III. If a print fails and you do not understand why, the answer is in Part I, Chapter 7. If you are preparing a manuscript or patent, Part IV is mandatory reading.

Contents

Conceptual Framework: DIW as a Coupled System	4
Part I – Theory	5
1 Why DIW is Not FFF	5
1.1 Three control modes, one printer family	6
2 Rheology in Depth	6

2.1	Stress, strain, and strain rate	7
2.2	Yield stress and the two-viscosity problem	7
2.3	Why DIW needs shear thinning	7
2.4	Two constitutive models you actually use	7
2.5	Viscoelasticity: G' and G''	8
2.6	Recovery: the time-domain check	9
2.7	Dimensionless rheological numbers	10
3	The Piston-Driven Hydraulic Loop	11
3.1	Mass balance: from V_p to flow	11
3.2	Velocity profile inside the needle	12
3.3	Mass balance: from V_p to print speed	12
4	From Flow to Pressure: Hagen–Poiseuille for Non-Newtonian Fluids	14
4.1	Newtonian baseline	14
4.2	Power-Law extension	14
4.3	Cross-model extension (Rabinowitsch–Weissenberg)	14
4.4	Which model to trust?	15
4.5	Wall shear rate and stress	15
5	Flow Factor, Die Swell, and the Slicer Translation	16
5.1	The crucial inversion: k_{flow} is not the number you type into the slicer	16
5.2	Die-swell proxy from Cross m	17
6	The Maximum Scaffold Height	18
6.1	Three heights, three scales	18
6.2	Deriving $h_{\text{scaffold}}^{\text{max}}$	18
6.3	What is σ_{max} ? Three criteria	19
6.4	Three numbers per ink	19
6.5	Lattices and unsupported spans – the Smay criterion	20
7	Failure Morphology: Physical Causes and Corrective Actions	21
7.1	Under-extrusion	21
7.2	Over-extrusion	22
7.3	Die swell excess	22
7.4	Scaffold collapse (base bulging)	22
7.5	Lattice strut buckling	23
7.6	Intermittent extrusion (motor skip)	23
7.7	Layer-by-layer fusion (loss of resolution)	23
7.8	Stringing during travel	23
7.9	Rough filament surface / extrudate distortion	24
	Part II – Inks and Instruments	24
8	The Three Project Inks	24
8.1	C15 – CMC baseline	24
8.2	C15 Gira 5.5 – CMC + sunflower-oil nanoemulsion (NE)	25
8.3	Bozano – commercial hair gel	25

9 Nanoemulsion Mechanics	26
9.1 Nanoemulsion structure and droplet characteristics	26
9.2 How the NE stiffens the CMC matrix	26
9.3 Emulsion stability under shear	27
9.4 Structural recovery enhancement mechanism	27
9.5 Carrier model rationale	27
10 Rheological Tests: Instrument and Battery	27
10.1 Instrument	27
10.2 Standard test battery	28
10.3 Interpreting pre-shear effects	28
10.4 Interpreting the SAOS output: a field guide	28
11 Extended Material Classes	29
11.1 Cellulose-based inks	29
11.2 Alginate-based inks	29
11.3 Gelatin-based inks	29
11.4 Composite systems: natural + synthetic	29
Part III – Protocols	30
12 Workflow Overview	30
13 Step 2: Python Analysis	30
13.1 Pre-run checklist	30
13.2 Flow-curve fitting – <code>Fit_Muitos_Modelos_v4.py</code>	30
13.3 Recovery – <code>Recovery_v1.py</code>	31
13.4 Scaffold-height extraction – <code>extract_hmax_v2.py</code>	31
14 Step 3: MATLAB Simulation – <code>run_solver_v3.m</code>	31
15 Step 4: Calibration-Cube Protocol (C1–C5)	32
15.1 C1 – print-speed sweep	32
15.2 C2 – mass-per-line (k_{flow}) calibration	32
15.3 C3 – layer-height sweep	32
15.4 C4 – vertical-tower test	32
15.5 C5 (optional) – bridging test for lattice scaffolds	33
16 Step 5: Final Scaffold Print	33
17 Troubleshooting: Failure-Mode Decision Tree	33
18 The Friday Print – Closing the Loop	33
Part IV – Extended Topics	34
19 Biological and Regulatory Considerations	34
19.1 Cell viability and wall shear stress	34
19.2 Plug flow and cell protection	35
19.3 Long-duration print considerations	35
19.4 MAPA-oriented regulatory considerations	35

20 Statistical Optimisation: DOE and Response Surface Methodology	36
20.1 Overview	36
20.2 Continuous vs. categorical variables	36
20.3 Model selection recommendations	37
20.4 Responses and their transformations	37
A Glossary	37
B Notation and Unit Reference	39
C Worked End-to-End Example: C15 Gira 5.5 on 21G	39
D Engineering Design Rules – Consolidated Reference	40
E Single-Page Run Sheet	41
F Software Setup Guide	41
F.1 Python environment	41
F.2 MATLAB	42
F.3 LaTeX	42

Conceptual Framework: DIW as a Coupled System

Direct Ink Writing is not a single-physics problem. It is a **coupled system** in which formulation chemistry, network rheology, extrusion hydraulics, deposition geometry, and scaffold mechanics interact simultaneously. Every printing parameter is both a cause and a consequence in this web. The central thesis of this textbook is: *understand the coupling, and every slicer entry follows by algebra*.

Figure 1 shows the complete coupling map.

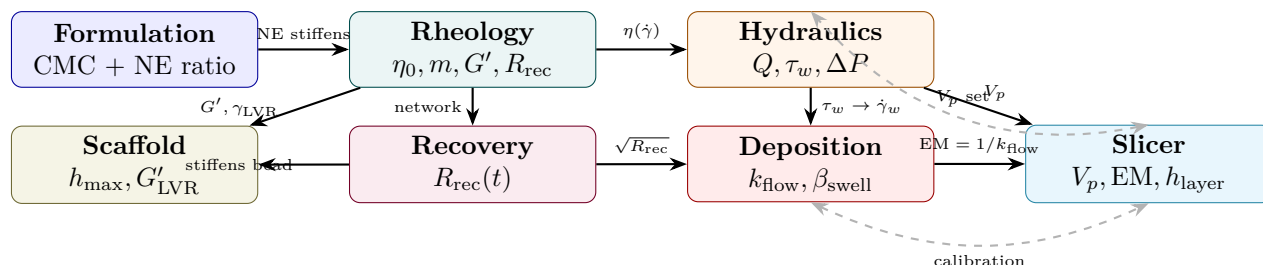


Figure 1: Systems-level coupling map for piston-driven DIW. Solid arrows: physical causation. Dashed: feedback via calibration or operator control. The operator directly commands only V_p (piston velocity); every other quantity is a consequence of the ink’s rheology and the system geometry.

PARAMETER INTERACTIONS

How parameters couple – the four most important interactions.

1. **Increasing V_p raises Q , which raises $\dot{\gamma}_w$, which lowers η (shear thinning), which raises τ_w sub-linearly.** Pressure does not scale linearly with V_p for shear-thinning inks: doubling V_p typically increases ΔP by a factor of 2^n ($n < 1$).
2. **Increasing V_p also raises v_{print} , which shortens dwell time per layer, which changes the SAOS criterion controlling $h_{\text{scaffold}}^{\text{max}}$.** Faster printing can mean the $G'(\omega = 1)$ criterion applies rather than the $G'(\omega = 0.01)$ one – apparently improving height

capacity.

3. **Adding NE raises η_0 and G'_{LVR} but slightly lowers R_{rec} .** The first two effects improve scaffold height and resist gravity; the last increases the required Extrusion Multiplier. Both directions must be tracked simultaneously.
4. **Narrower needle reduces R_n , massively raising ΔP ($\propto R_n^{-4}$) and $\dot{\gamma}_w$ ($\propto R_n^{-3}$) at the same V_p .** Switching from 21G to 22G at fixed V_p raises wall shear stress and may breach cell-safety thresholds even though the ink parameters are unchanged.

RUNNING EXAMPLE – The Friday print

Monday, 9:00. You are Ana, a new intern at BioPoli. Your supervisor hands you a sealed jar of *C15 Gira 5.5* (CMC + sunflower-oil nanoemulsion ink, the **headline ink** of this project), a 21G blunt needle, and the following brief:

“By Friday at 16:00 I need a printed solid block, roughly $1 \times 1 \times 2 \text{ cm}^3$, of this ink. The shape doesn’t matter – any rectangular prism is fine. The Wednesday morning slot on the printer is yours. Don’t waste ink.”

Your job between now and Wednesday: extract, from the analysis scripts and this book, every number you must type into the slicer. Wednesday: calibrate at the bench. Thursday: print. Friday: document and deliver.

By the end of this book you will have answered, in order:

1. **Can this ink print at all?** (Ch. 2)
2. **What piston velocity V_p should I command?** (Ch. 3)
3. **Is the resulting extrusion pressure safe?** (Ch. 4)
4. **What flow factor k_{flow} (and slicer Extrusion Multiplier) do I enter?** (Ch. 5)
5. **Will a 20 mm-tall block survive its own weight?** (Ch. 6)
6. **What does my Wednesday calibration morning look like?** (Ch. 15)

After each chapter above, you will find a box titled “Running Example – Solution to QN”, solving one question with explicit numbers and marking what is still unknown.

Part I – Theory

1 Why DIW is Not FFF

Direct Ink Writing (DIW) extrudes a viscous fluid through a needle and relies on the fluid’s own rheology to retain its shape after deposition. Fused Filament Fabrication (FFF) melts a thermoplastic, extrudes it as a controlled-diameter molten strand, and solidifies it on contact with the cooler build plate. The two processes look similar (an extruder moves over a build plate), but the physics is fundamentally different, and the slicer firmware – written for FFF – does *not* translate cleanly.

PHYSICAL INTUITION

In FFF, the filament has a fixed diameter (1.75 or 2.85 mm). The extruder pushes a known length of filament, giving a known volume of melt. “How much material was deposited?” is answered by E-axis geometry alone.

In DIW, the ink is a free fluid in a syringe. “How much material was deposited?” depends on: (i) how fast the piston moves; (ii) how the ink responds to shear in the needle; (iii) how the deposited filament swells or relaxes after leaving the needle; (iv) how the network recovers in the time between deposition and the next layer.

The slicer knows none of that. It still thinks it is FFF. Our job is to compute the right numbers *outside* the slicer and feed them in as if they were FFF parameters.

1.1 Three control modes, one printer family

DIW printers differ in what the operator commands:

1. **Pneumatic DIW.** Operator commands ΔP (the gas pressure driving the syringe). Flow rate Q is the consequence. Dominant in academic literature; simpler hardware and lower cell stress at constant pressure.
2. **Screw / progressive-cavity DIW.** Operator commands auger rotation, which translates to a quasi-constant Q . Robust against air bubbles; more expensive; common in commercial bioprinting.
3. **Piston / volumetric DIW.** Operator commands piston linear velocity V_p (via a stepper-motor lead screw). Volumetric flow Q is fixed by syringe geometry. ΔP is whatever the rheology demands. **This is the lab’s printer.**

CONTROL PARADIGM

The lab DIW printer is **piston-driven / volumetric**. The single flow input is the piston velocity V_p (mm/s), entered into the slicer firmware as the E-axis feedrate. Everything downstream – Q , $\bar{v}_n$, τ_w , ΔP , v_{print} – follows from V_p and the ink rheology. They are **outputs and diagnostics**, not knobs.

Implication for literature reading. A paper that says “we extruded at 100 kPa” must be translated to a V_p on our printer before the number is useful. The translation goes through the rheological model (Ch. 4).

KEY TAKEAWAYS

- DIW is a rheology-dependent process; FFF slicers must be “tricked” with pre-computed parameters.
- On the lab printer, V_p is the only flow control input. ΔP is a diagnostic, never a knob.
- Comparing across literature requires knowing the control mode (pneumatic, screw, or piston).

ENGINEERING DESIGN RULES

DR 1.1. Before reading any DIW paper, identify the control mode. Pneumatic data give pressure–flowrate relationships; piston data give velocity–flowrate relationships. They are not interchangeable without going through the constitutive model.

DR 1.2. Never enter ΔP directly into the slicer – it does not exist as a slicer input on this printer. If a calibration strategy asks for a pressure setting, it was written for pneumatic DIW.

2 Rheology in Depth

Rheology is the science of how matter deforms and flows. For DIW we need seven concepts: stress, strain rate, viscosity, shear thinning, yield stress, viscoelasticity, and structural recovery.

2.1 Stress, strain, and strain rate

Place a fluid between two parallel plates separated by a gap h . Hold the bottom plate fixed; slide the top plate at velocity V under a tangential force F over area A . The fluid is subjected to:

- a **shear stress** $\tau = F/A$, in pascals (Pa),
- a **shear strain rate** $\dot{\gamma} = V/h$, in inverse seconds (s^{-1}).

For a Newtonian fluid the two are linearly related:

$$\tau = \eta \dot{\gamma} \quad (1)$$

where η (Pa s) is the dynamic viscosity. Water has $\eta \approx 10^{-3}$ Pa s; honey, $\eta \approx 10$ Pa s; our CMC inks span $\eta \sim 100$ Pa s at high shear to $\eta \sim 10^3$ Pa s at rest – four orders of magnitude.

2.2 Yield stress and the two-viscosity problem

Many structured fluids behave like solids at low stress and like viscous fluids above a critical threshold called the **yield stress** τ_y . The simplest model is the Bingham fluid:

$$\tau = \tau_y + \eta_p \dot{\gamma} \quad (\tau > \tau_y); \quad \dot{\gamma} = 0 \quad (\tau \leq \tau_y). \quad (2)$$

For DIW, a non-zero τ_y is advantageous: once extruded, the material stops flowing at rest and holds its shape. The Herschel-Bulkley model generalises this to power-law viscosity above yield:

$$\tau = \tau_y + K \dot{\gamma}^n \quad (\tau > \tau_y). \quad (3)$$

In practice, the inks in this project behave as *apparent* yield stress fluids: the zero-shear plateau of the Cross model (η_0) is so high that the material appears solid on lab timescales. We do not formally fit τ_y but we extract an equivalent from SAOS data (Ch. 6).

2.3 Why DIW needs shear thinning

If our inks were Newtonian, we would face a brutal compromise: a viscosity high enough to keep the printed filament from slumping (say, 1000 Pa s) would require extrusion pressures orders of magnitude above any safe syringe rating. The only way out is to design an ink whose viscosity drops under shear and recovers at rest:

$$\eta(\dot{\gamma}) = \text{decreasing function of } \dot{\gamma}. \quad (4)$$

PHYSICAL INTUITION

A good DIW ink has two effective viscosities:

- a *low* in-needle viscosity, where $\dot{\gamma} \sim 10\text{--}10^3 \text{ s}^{-1}$, so ΔP stays manageable;
- a *high* post-deposition viscosity, where $\dot{\gamma} \rightarrow 0$, so the filament resists gravity.

The printability problem reduces to: *does the ink switch fast enough between these two regimes, and does it recover fast enough to hold successive layers?*

2.4 Two constitutive models you actually use

The viscosity-shear-rate curve $\eta(\dot{\gamma})$ is fitted to a mathematical form so it can be plugged into pressure-drop calculations. The two models used in this project are the Power Law and the Cross model.

Power Law.

$$\eta(\dot{\gamma}) = K \dot{\gamma}^{n-1}, \quad \tau = K \dot{\gamma}^n. \quad (5)$$

Two parameters: K (Pa s^n) sets the viscosity scale; n ($0 < n < 1$ for shear thinning) controls how rapidly the viscosity drops. Newtonian is recovered at $n = 1$.

Virtues: analytical extrusion solutions (Eq. 15), two-parameter simplicity, easy to invert.

Limitations: predicts $\eta \rightarrow \infty$ at $\dot{\gamma} \rightarrow 0$ and $\eta \rightarrow 0$ at $\dot{\gamma} \rightarrow \infty$ – both unphysical. Valid only within a single decade of the fitted $\dot{\gamma}$ range.

Cross model.

$$\eta(\dot{\gamma}) = \eta_\infty + \frac{\eta_0 - \eta_\infty}{1 + (\lambda \dot{\gamma})^m}. \quad (6)$$

Four parameters: η_0 (zero-shear rest viscosity), η_∞ (infinite-shear viscosity), λ (transition time constant – $1/\lambda$ is the centre-of-transition shear rate), and m (the sharpness exponent, $0 < m \leq 1$).

Virtues: finite rest and high-shear viscosities, smooth transition, physically motivated, extends naturally across many decades of $\dot{\gamma}$.

Limitations: four parameters increase fitting sensitivity; η_∞ is often poorly constrained at high $\dot{\gamma}$.

Interpreting m physically. $m \rightarrow 1$ means the ink transitions sharply from η_0 to η_∞ – almost a binary switch. $m \approx 0.5$ means a gradual decline. A high m implies the ink stores very little elastic energy during shear-thinning (less die swell on exit). A low m implies broader energy storage and larger die swell.

WORKED EXAMPLE – the three inks

Cross parameters for the three inks (Ramp 1, no pre-shear, Fit_Muitos_Modelos_v4):

Ink	η_0 [Pa s]	η_∞ [Pa s]	λ [s]	m [-]	R^2
C15	578.3	0.0163	3.71	0.811	0.9956
C15 Gira 5.5	2439.5	0.001	4.10	0.988	0.9882
Bozano	6354.6	0.932	73.05	0.866	0.9937

Reading: η_0 ranks Bozano $\gg$ NE $\gg$ C15 at rest. The NE adds roughly a factor of four to the base CMC. Bozano is in a different class. λ tells you where the transition is centred: $1/\lambda \approx 0.25 \text{ s}^{-1}$ for C15 and NE vs. $\approx 0.014 \text{ s}^{-1}$ for Bozano. $m = 0.988$ for the NE means an almost-binary switch – minimal elastic energy storage – hence minimal die swell.

2.5 Viscoelasticity: G' and G''

The flow curve tells us how the ink behaves under *continuous* shear. But during printing, deposition is followed by rest, then by the next layer; the ink must also respond well under *small, oscillating* strains. That is the domain of small-amplitude oscillatory shear (SAOS).

Apply a sinusoidal strain $\gamma(t) = \gamma_0 \sin(\omega t)$ at fixed amplitude γ_0 and angular frequency ω . The resulting stress is also sinusoidal but phase-shifted:

$$\tau(t) = \gamma_0 [G'(\omega) \sin \omega t + G''(\omega) \cos \omega t]. \quad (7)$$

We extract two moduli:

- $G'(\omega)$ – the **storage modulus**: in-phase with strain, captures the elastic / solid-like part.
- $G''(\omega)$ – the **loss modulus**: $\pi/2$ out of phase, captures the viscous / liquid-like part.

The ratio $\tan \delta = G''/G'$ (loss tangent) summarises the character: $\tan \delta < 1$ means gel-like; $\tan \delta > 1$ means liquid-like.

Connecting G'/G'' to print outcomes.

Rheological feature	What it predicts	Failure if violated
High G'_{LVR}	Filament stiff after deposition	Layer collapse; gravity-driven sag
Wide γ_{LVR}	Tolerant to deformation	None – wider is better
Flat $G'(\omega)$ (true gel)	Stable across all print timescales	Quasi-static collapse on slow prints
Sloped $G'(\omega)$ (critical gel)	Modulus drops at long timescale	Collapse during slow or paused prints
High G'' / low $\tan \delta$	Energy dissipation after deposition	Elastic rebound, bouncing filament

Two SAOS tests, two pieces of information.

1. **Amplitude sweep.** Fix ω , sweep γ_0 small to large. At small strain, G' is constant – the *linear viscoelastic regime* (LVR). At the critical strain γ_{LVR} (defined where G' has dropped 10% from its plateau), the network is permanently damaged. Beyond γ_{LVR} the response is nonlinear.
2. **Frequency sweep.** Fix γ_0 inside the LVR, sweep ω . $G'(\omega)$ encodes the network's timescale response: high ω probes fast (in-needle) dynamics; low ω probes slow (post-deposition, layer-stacking) dynamics.

WORKED EXAMPLE – the three inks

SAOS-derived numbers for the three inks:

Ink	G'_{LVR} [Pa]	γ_{LVR} [%]	$G'(\omega = 1 \text{ rad s}^{-1})$ [Pa]
C15	270	64.0	114
C15 Gira 5.5	916	60.9	228
Bozano	348	13.7	346

Reading: Adding NE to C15 raises G'_{LVR} from 270 to 916 Pa ($3.4\times$ stiffening) without narrowing the LVR (61% vs. 64%). The NE acts as a *reinforcing filler*, not a crosslinker. A crosslinker would narrow the LVR. Bozano has a comparable absolute modulus at $\omega = 1$ (346 Pa) but a far narrower LVR (13.7%): it is stiff but brittle to shear deformation.

2.6 Recovery: the time-domain check

The amplitude sweep tells us *when* the network breaks; the recovery test tells us *how fast it heals*. Protocol (three-interval thixotropy test, `Recovery_v1.py`):

1. **Interval 1 (rest reference):** low shear $\dot{\gamma}_1 = 0.1 \text{ s}^{-1}$; record η_1 .
2. **Interval 2 (disruption):** high shear $\dot{\gamma}_2 \in \{50, 100, 150\} \text{ s}^{-1}$ (matched to the needle wall shear rate of the intended print), 60 s. Network is broken.
3. **Interval 3 (recovery):** return to $\dot{\gamma}_1$. Watch $\eta(t)$ rebuild. Define $R_{\text{rec}}(\%) = \eta_3(t = 30 \text{ s})/\eta_1 \times 100$.

Higher R_{rec} means the network heals faster, reducing sag of the deposited filament before the next layer arrives.

Thixotropy vs. rheopexy. Compare the fitted consistency index K from the Flow (no pre-shear) and Flow Sheared curves:

- $K_{\text{sheared}} < K$: structure broken by pre-shear, recovers incompletely $\Rightarrow$ **thixotropic**.

- $K_{\text{sheared}} > K$: pre-shear has organised the network $\Rightarrow$ **rheopexic** (shear-induced structuring).

This distinction is mechanistically significant and must be discussed explicitly in manuscripts.

WORKED EXAMPLE – the three inks

Ink	R_{rec} at 50 s^{-1}	100 s^{-1}	150 s^{-1}
C15	75%	60%	55%
C15 Gira 5.5	60%	45%	40%
Bozano	99%	98%	96%

Bozano recovers almost completely at every shear rate – consistent with its true-gel character. The NE recovers *less* than pure C15: the NE stiffens the rest network (higher G') but the network bonds are slightly more brittle to high-shear disruption. This trade-off must be compensated at the slicer with a higher Extrusion Multiplier.

2.7 Dimensionless rheological numbers

Three dimensionless groups govern whether a DIW ink will print, store shape, and survive long prints.

DIMENSIONLESS ANALYSIS

Deborah number $De = t_{\text{relax}}/t_{\text{process}}$: ratio of the material's relaxation time to the characteristic process time. For SAOS: $De = \lambda\omega$. If $De \gg 1$, the material is elastic at that timescale; if $De \ll 1$, it is viscous.

- In the needle, $t_{\text{process}} \sim R_n/\bar{v}_n \sim \text{ms}$: $De \gg 1$ for all three inks. Elastic behaviour dominates $\Rightarrow$ die swell is expected.
- After deposition, $t_{\text{process}} \sim \text{layer-dwell time}$ (1–100 s): De decreases. If $De < 1$ at this timescale, the material has relaxed and lost memory – it is susceptible to sag.

Bingham number $Bi = \tau_y/(\eta_p \dot{\gamma}_w)$ (for yield-stress fluids, or analogously $\eta_0/(\eta_\infty \cdot (\lambda\dot{\gamma}_w)^m)$ for Cross): measures whether viscosity or yield stress dominates in the needle.

- $Bi \gg 1$: plug flow in the centre with thin shear layers at the wall (preferred for cell viability – most cells are in the low-shear core).
- $Bi \ll 1$: Newtonian-like parabolic profile.

Capillary number $Ca = \eta \dot{\gamma}/(2\sigma/R)$: ratio of viscous to surface tension forces at the needle exit. If $Ca \gg 1$, the filament is controlled by viscosity and recovers slowly; die swell is purely elastic in origin. If $Ca \sim 1$, surface tension is non-negligible and affects bead shape on the substrate. For hydrogels ($\sigma \sim 0.07 \text{ N m}^{-1}$) and $R \sim 0.26 \text{ mm}$, $Ca \gg 1$ in the printable $\dot{\gamma}$ range – surface tension is secondary.

RUNNING EXAMPLE – Solution to Q1

Question. Can the C15 Gira 5.5 ink print at all?

Answer. Ana reads the parameter tables above and notes three signals:

1. Cross $m = 0.988$. Nearly binary shear thinning: $\eta_0 \approx 2440 \text{ Pa s}$ at rest; collapses sharply above $\dot{\gamma} \approx 0.25 \text{ s}^{-1}$. *Excellent printability signature.*
2. $\gamma_{\text{LVR}} = 60.9\%$. Wide linear regime: the deposited filament tolerates up to 60% strain before yielding. *Tolerant to small perturbations during deposition.*

3. $R_{\text{rec}} \approx 45\%$ at 100 s^{-1} . Each layer regains roughly half its rest modulus within 30 s.
Manageable – compensated by the Extrusion Multiplier.

Verdict. Rheologically well-suited to DIW. Proceed.

Still unknown: what V_p to enter into the slicer. Q2.

KEY TAKEAWAYS

- Shear thinning is *necessary* for DIW: low in-needle viscosity, high post-deposition viscosity.
- The Cross model ($\eta_0, \eta_\infty, \lambda, m$) is preferred over the Power Law for the full $\dot{\gamma}$ range.
- G'_{LVR} and γ_{LVR} from SAOS predict scaffold height and tolerance to deformation.
- Recovery R_{rec} directly feeds the deposition efficiency k_{flow} and the required Extrusion Multiplier.
- Dimensionless numbers (De, Bi) explain velocity profiles inside the needle and post-deposition stability.

ENGINEERING DESIGN RULES

DR 2.1. Always report both G'_{LVR} and γ_{LVR} – neither alone is sufficient. $G'_{\text{LVR}} \cdot \gamma_{\text{LVR}}$ is the yield-stress proxy that controls scaffold height.

DR 2.2. Prefer the Cross model for printability prediction. Only use Power Law for single-decade fitting when analytical expressions are needed and the Cross fit is poorly constrained.

DR 2.3. Structural recovery is more important than static viscosity for scaffold quality. An ink that recovers slowly will produce under-built beads even if η_0 is high.

DR 2.4. When $m \approx 1$ (NE), expect minimal die swell and minimal β_{swell} . When $m < 0.7$, measure die swell directly in calibration test C1.

3 The Piston-Driven Hydraulic Loop

We now connect the ink rheology to the printer hardware. The system is a simple hydraulic circuit: piston → syringe barrel → conical transition → needle.

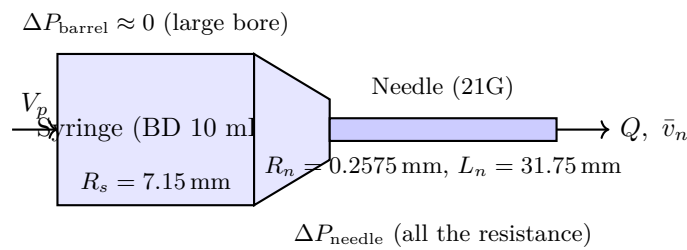


Figure 2: The DIW hydraulic loop. The operator commands V_p ; syringe geometry fixes Q ; the ink rheology and needle geometry fix τ_w and ΔP . The syringe barrel contributes negligible pressure drop because $(R_s/R_n)^4 \approx 5 \times 10^5$ – almost all resistance is in the needle.

3.1 Mass balance: from V_p to flow

The piston advances at V_p , displacing fluid at volumetric rate

$$Q = \pi R_s^2 V_p. \quad (8)$$

The same Q passes through the needle (continuity):

$$\bar{v}_n = \frac{Q}{\pi R_n^2} = \left(\frac{R_s}{R_n} \right)^2 V_p. \quad (9)$$

For our hardware ($R_s = 7.15$ mm, 21G $R_n = 0.2575$ mm), the area ratio $(R_s/R_n)^2 \approx 771$. A 1 mm/s **piston velocity becomes a 771 mm/s mean needle velocity**.

3.2 Velocity profile inside the needle

The shape of the velocity profile inside the needle depends on the Bingham number. Figure 3 shows the two limits.

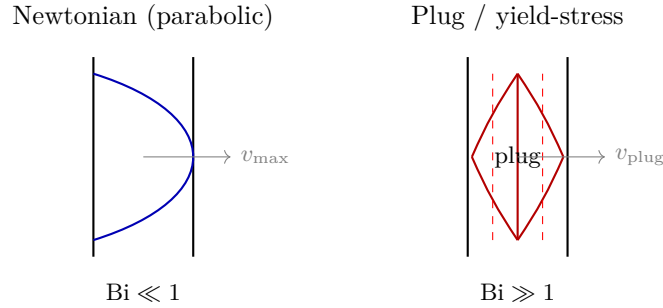


Figure 3: Velocity profiles in the needle cross-section. Left: Newtonian / power-law fluid – maximum shear at the wall, zero at the centreline. Right: plug-flow limit for a high-yield-stress fluid – most of the cross-section moves as a solid plug; shear is concentrated in thin wall layers. Our CMC inks lie between these limits; high-modulus inks (Bozano) approach the plug-flow regime.

Biological significance. Cells suspended in the ink experience the *local* shear rate, not the wall average. In plug flow, cells in the core see very low shear stress. For bioprinting with live cells, inks with partial plug-flow behaviour (high Bi) are preferable because the cell-damaging high-shear zone is confined to a thin wall layer.

3.3 Mass balance: from V_p to print speed

The print head deposits a filament of cross-section $w h_{\text{layer}}$ as it moves over the build plate at speed v_{print} :

$$Q = v_{\text{print}} w h_{\text{layer}}. \quad (10)$$

With $w \approx 2R_n(1 + \beta_{\text{swell}})$ and $h_{\text{layer}} = h_{\text{factor}} \cdot 2R_n$ this becomes

$$v_{\text{print}} \approx \frac{\pi(R_s/R_n)^2}{4 h_{\text{factor}}(1 + \beta_{\text{swell}})} V_p \approx \frac{700-865}{(1 + \beta_{\text{swell}})} V_p \quad (21\text{G}). \quad (11)$$

PHYSICAL INTUITION

The piston is the slowest thing. On 21G, V_p in mm/s maps to v_{print} in mm/s with a multiplier of $\approx 700-865$:

- $V_p = 0.005 \Rightarrow v_{\text{print}} \approx 4$ mm/s
- $V_p = 0.010 \Rightarrow v_{\text{print}} \approx 8$ mm/s
- $V_p = 0.030 \Rightarrow v_{\text{print}} \approx 25$ mm/s
- $V_p = 0.050 \Rightarrow v_{\text{print}} \approx 40$ mm/s

The mechanical envelope of the lab gantry is roughly $v_{\text{print}} \in [3, 30]$ mm/s, so $V_p \in [\sim 0.004, \sim 0.04]$ mm/s.

DIMENSIONLESS ANALYSIS

Generalised Reynolds number for power-law flow in a tube:

$$\text{Re}_{\text{gen}} = \frac{\rho \bar{v}_n^{2-n} (2R_n)^n}{K 8^{n-1}} \left(\frac{4n}{3n+1} \right)^n. \quad (12)$$

For laminar flow (the fundamental assumption of all our ΔP equations), $\text{Re}_{\text{gen}} \ll 2100$. For our operating conditions:

Ink (PL fit)	K [Pas ^{n}]	Re_{gen} at $V_p = 0.01$ mm/s
C15	78	$\ll 1$ (deeply laminar)
C15 Gira 5.5	267	$\ll 1$
Bozano	1086	$\ll 1$

All three inks are deeply laminar across the entire printable V_p range. The fully-developed laminar flow assumption is valid. No turbulence correction is needed.

RUNNING EXAMPLE – Solution to Q2

Question. What piston velocity V_p should Ana command?

Reasoning. The mechanical envelope is $v_{\text{print}} \in [3, 30]$ mm/s. Ana picks a target of $v_{\text{print}} \approx 8$ mm/s – slow enough to forgive minor mis-calibration, fast enough to finish the production print in under an hour.

For 21G with $\beta_{\text{swell}} \approx 0$ (justified for the NE, see Q4), Eq. 11 gives amplification ≈ 865 :

$$V_p = \frac{8}{865} \approx 0.0093 \text{ mm/s} \rightarrow \text{rounded to } \mathbf{0.01} \text{ mm/s.}$$

Sanity check. $Q = \pi(7.15)^2 \cdot 0.01 = 1.6 \text{ mm}^3/\text{s} \approx 100 \mu\text{L}/\text{min}$ – typical for syringe bioprinting.

Still unknown: is the resulting ΔP within hardware limits? Q3.

KEY TAKEAWAYS

- $V_p \rightarrow Q \rightarrow \bar{v}_n$: a factor of ~ 771 velocity amplification on 21G. The piston is slow; the needle is fast.
- $\text{Re}_{\text{gen}} \ll 2100$ for all printable conditions: laminar flow is always valid.
- Plug flow in high-Bi inks protects cells in the core from damaging shear.
- v_{print} is set by V_p and geometry – it is not an independent knob unless slicer X/Y feedrate is decoupled from E.

ENGINEERING DESIGN RULES

DR 3.1. Always verify $\text{Re}_{\text{gen}} \ll 2100$ when changing needle gauge or ink. Turbulent transition would invalidate all ΔP and τ_w calculations.

DR 3.2. Do not change needle gauge (R_n) without recomputing all downstream quantities – the R_n^{-4} scaling of ΔP makes gauge changes the largest single-factor lever in the whole workflow.

DR 3.3. For cell-loaded inks, prefer inks with high Bi (approaching plug flow) to confine high shear to thin wall layers.

4 From Flow to Pressure: Hagen–Poiseuille for Non-Newtonian Fluids

Given Q , we need ΔP across the needle – both as a diagnostic (is the print mechanically feasible?) and as a downstream input to wall-shear calculations.

4.1 Newtonian baseline

For a Newtonian fluid in a straight cylinder of radius R and length L , Hagen–Poiseuille gives

$$\Delta P = \frac{8\eta L Q}{\pi R^4}. \quad (13)$$

The R^{-4} dependence is severe: halving the needle radius multiplies the pressure 16-fold. This is the dominant reason needle gauge matters so much.

4.2 Power-Law extension

For a Power-Law fluid, the analytical solution for steady laminar fully- developed flow in a cylinder gives wall shear stress:

$$\tau_w = K \left(\frac{(3n+1) Q}{\pi R^3 n} \right)^n, \quad (14)$$

and pressure drop from a force balance:

$$\Delta P = \frac{2L\tau_w}{R}. \quad (15)$$

DERIVATION

A force balance on a fluid cylinder of radius r inside the needle: net axial pressure force $\pi r^2 \Delta P$ equals integrated shear over its sides $2\pi r L \tau(r)$. Therefore $\tau(r) = (\Delta P/2L) r$ – shear stress is *linear* in r , peaking at $\tau_w = \Delta P R/2L$ at the wall.

For a Power-Law fluid $\tau = K \dot{\gamma}^n$, the velocity profile $v(r)$ follows by integration; integrating $v(r)$ over the cross-section gives Q as a function of τ_w – inverting yields Eq. 14.

Physical consequence. The linear $\tau(r)$ profile is universal – it holds regardless of the constitutive model (Newtonian, PL, Cross, Bingham). Only the velocity profile $v(r)$ changes with the model.

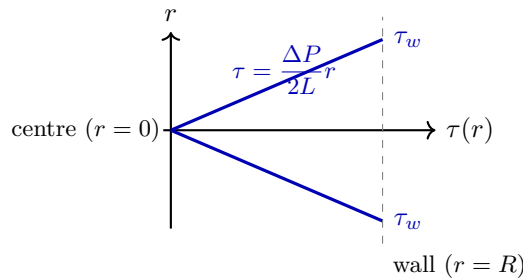


Figure 4: Shear stress profile inside the needle. The stress is always linear in r , peaking at the wall with value $\tau_w = \Delta P R/2L$ and zero at the centreline. This universal result holds for any constitutive model.

4.3 Cross-model extension (Rabinowitsch–Weissenberg)

The Cross model has no closed-form $Q(\tau_w)$ expression. We use the Rabinowitsch–Weissenberg integral:

$$Q(\tau_w) = \frac{\pi R^3}{\tau_w^3} \int_0^{\tau_w} \tau^2 \dot{\gamma}(\tau) d\tau, \quad (16)$$

where $\dot{\gamma}(\tau)$ is obtained by inverting Eq. 6 (solve $\eta(\dot{\gamma})\dot{\gamma} = \tau$ for $\dot{\gamma}$ numerically – a single `fzero` call per point). The MATLAB code at `bioprinting_algorithm_v3.m` runs this on a 400-point grid for each V_p .

4.4 Which model to trust?

Both models are available in the MATLAB output, and they are compared side-by-side.

- **In the printable range** ($\dot{\gamma}_w \sim 50\text{--}1000\text{ s}^{-1}$ for 21G), both agree within 20% for our inks.
- **At low $\dot{\gamma}_w$** (very slow prints or 22G at low V_p): the Power Law over-estimates ΔP because it has no η_0 plateau. The Cross model is correct.
- **Preferred:** Cross for all production runs. Power Law reserved for quick back-of-envelope estimates.

4.5 Wall shear rate and stress

Once τ_w is known, wall shear rate follows by inverting the constitutive model:

$$\dot{\gamma}_w = (\tau_w/K)^{1/n} \text{ (PL); } \quad \dot{\gamma}_w \text{ by } \texttt{fzero} \text{ on } \eta(\dot{\gamma})\dot{\gamma} = \tau_w \text{ (Cross)}. \quad (17)$$

$\dot{\gamma}_w$ is the shear rate cells see at the needle wall, and is the disruption shear rate to use when evaluating the recovery test result.

Cell-safety threshold. Literature consensus places the upper bound for human cells at $\tau_w \sim 5\text{ kPa}$ (?). Above this, cell viability drops sharply. Both τ_w and cumulative shear exposure (Ch. 19) must be checked for cell-loaded prints.

WORKED EXAMPLE – the three inks

Hagen–Poiseuille on 21G at $V_p = 0.01\text{ mm/s}$ (Cross model):

Ink	Q [mm ³ /s]	τ_w [Pa]	$\dot{\gamma}_w$ [s ⁻¹]	ΔP [kPa]
C15	1.61	564	220	139
C15 Gira 5.5	1.61	663	60	164
Bozano	1.61	469	8	116

All three inks deliver the same Q (geometry-fixed), but at very different τ_w and $\dot{\gamma}_w$ – the rheology decides. Pressures sit at 100–170 kPa, well below the 1500 kPa syringe ceiling and the 5 kPa cell-safe wall-shear ceiling.

RUNNING EXAMPLE – Solution to Q3

Question. At $V_p = 0.01\text{ mm/s}$ on 21G, is the extrusion pressure safe for C15 Gira 5.5?

From the worked example above:

$$\Delta P_{\text{Cross}} \approx 164\text{ kPa}, \quad \tau_w \approx 663\text{ Pa}, \quad \dot{\gamma}_w \approx 60\text{ s}^{-1}.$$

Safety check:

- Syringe ceiling 1500 kPa: $164/1500 = 11\%$. Comfortable.
- Cell-safety $\tau_w < 5\text{ kPa}$: 663 Pa is far below.
- Motor stall (estimated $\sim 500\text{--}800\text{ kPa}$): comfortable.

Verdict. Safe to print. At $V_p = 0.03\text{ mm/s}$, ΔP rises to $\sim 400\text{ kPa}$ (scaling as $V_p^{0.24}$ for $n = 0.24$) – still safe but with smaller margin.

Still unknown: the slicer Extrusion Multiplier. Q4.

KEY TAKEAWAYS

- The shear stress profile $\tau(r) = (\Delta P/2L)r$ is universal and linear – wall stress is always the maximum.
- $\Delta P \propto R^{-4}$: needle gauge is the dominant lever. Cross-model prediction is preferred over Power Law for all multi-decade ranges.
- $\tau_w < 5 \text{ kPa}$ is the cell-safety constraint. All three inks satisfy it comfortably at the nominal V_p .

ENGINEERING DESIGN RULES

DR 4.1. Always report ΔP , τ_w , and $\dot{\gamma}_w$ as a trio when characterising a print condition. ΔP alone is insufficient – τ_w determines cell survival.

DR 4.2. Use the Cross model for ΔP prediction. Report both Cross and Power Law outputs so inconsistencies can be flagged.

DR 4.3. For any new ink or needle combination, always run Re_{gen} and Bi estimates *before* the first print to confirm laminar flow and to anticipate the velocity profile type.

5 Flow Factor, Die Swell, and the Slicer Translation

The slicer treats the printer as FFF. It multiplies the commanded volumetric flow by an Extrusion Multiplier $\text{EM}_{\text{slicer}}$, which in FFF normally sits near 1.0. In DIW, three independent physical deviations change the effective deposited volume:

1. **Die swell** (β_{swell}): after leaving the needle, viscoelastic ink expands radially as stored elastic stresses relax. The deposited filament is wider than the needle ID. $w_{\text{line}}/\text{ID} = 1 + \beta_{\text{swell}}$.
2. **Syringe-wall slip** ($f_{\text{slip}} \leq 1$): imperfect piston sealing or ink-wall slip reduces the actual volume leaving the needle. Default 1.0 until C2 calibration.
3. **Viscoelastic recovery penalty** ($\sqrt{R_{\text{rec}}}$): if the network has only partially recovered when the next layer arrives, the deposited bead has not stiffened back up; material spreads and effective cross-sectional volume is lost.

$$k_{\text{flow}} = (1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}. \quad (18)$$

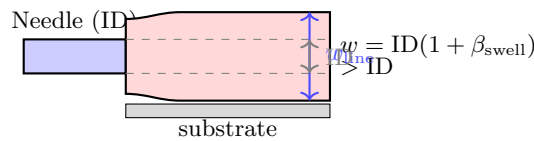


Figure 5: Die swell: the deposited filament is wider than the needle ID because elastic stresses stored during confinement in the needle relax on exit. Inks with $m \rightarrow 1$ (narrow transition) store little elastic energy and swell minimally. Inks with $m < 0.7$ may swell by $\beta > 0.15$ and require direct measurement in calibration test C1.

5.1 The crucial inversion: k_{flow} is not the number you type into the slicer

k_{flow} (Eq. 18) is the **deposition efficiency**: the fraction of the slicer-commanded volume that ends up locked in the bead:

$$k_{\text{flow}} = \frac{\text{volume locked in bead}}{\text{volume slicer was told to deliver}}. \quad (19)$$

The slicer’s Extrusion Multiplier is the *control knob* the operator uses to compensate for this inefficiency:

$$\text{EM}_{\text{slicer}} = \frac{1}{k_{\text{flow}}} = \frac{1}{(1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}}. \quad (20)$$

SAFETY / DATA INTEGRITY

The single most common operational error in DIW. Reading $k_{\text{flow}} = 0.67$ from the CSV and typing 0.67 into the slicer *worsens* under-build by another factor of $1/0.67 \approx 1.5$. The slicer multiplier is **always the inverse**: $\text{EM} = 1/k_{\text{flow}}$. When in doubt, verify from first principles: if the bead is too thin (under-extrusion), EM must go *up*.

Physical interpretation:

- $k_{\text{flow}} < 1$: bead loses material to swell-then-spread or recovery-limited slump. Slicer under-builds at nominal volume; raise EM above 1.
- $k_{\text{flow}} > 1$: bead locks in more volume than nominal (high recovery + large swell combine). Slicer over-builds; lower EM below 1.
- $k_{\text{flow}} = 1$: perfect FFF analogy; $\text{EM} = 1$.

5.2 Die-swell proxy from Cross m

Die swell is driven by stored elastic energy. An ink with $m \rightarrow 1$ (sharp transition) stores very little elastic energy during shear-thinning – hence minimal swell. Empirical proxy:

$$\beta_{\text{swell}} \approx 0.30(1 - m). \quad (21)$$

This spans $\beta = 0$ ($m = 1$, no swell) to $\beta = 0.30$ ($m = 0$, 30% swell). Direct measurement in C1 is preferred when $m < 0.7$.

WORKED EXAMPLE – the three inks

Predicted deposition efficiency and slicer Extrusion Multiplier ($f_{\text{slip}} = 1$, R_{rec} at $\dot{\gamma}_w \sim 100 \text{ s}^{-1}$):

Ink	m	β_{swell}	R_{rec} [%]	k_{flow}	$\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$
C15	0.811	0.057	60	0.86	1.17
C15 Gira 5.5	0.988	0.004	45	0.67	1.49
Bozano	0.866	0.040	98	1.07	0.93

The NE is the most demanding ink because of its low recovery: it requires $\sim 49\%$ more material at the slicer than the nominal FFF command. Bozano, with near-full recovery, actually asks for a $\sim 7\%$ reduction.

RUNNING EXAMPLE – Solution to Q4

Question. What Extrusion Multiplier should Ana enter for C15 Gira 5.5?

Step 1 – compute k_{flow} . $\beta_{\text{swell}} = 0.30(1 - 0.988) = 0.004$. $f_{\text{slip}} = 1.0$. $R_{\text{rec}} \approx 45\%$ at $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$:

$$k_{\text{flow}} = (1.004)^2 \cdot 1.0 \cdot \sqrt{0.45} \approx 0.67.$$

Step 2 – invert. $\text{EM}_{\text{slicer}} = 1/0.67 \approx \mathbf{1.49}$.

Why $\text{EM} > 1$? The NE has the lowest recovery (45%) of the three inks. Material leaves the needle but does not stiffen quickly enough before the next layer arrives; it spreads and loses effective cross-sectional volume. Pushing 49% more compensates.

Still unknown: will a 20-mm block survive its own weight? Q5.

KEY TAKEAWAYS

- k_{flow} is a *diagnostic* quantity (deposition efficiency); $\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$ is the number that goes into the slicer – never k_{flow} itself.
- Three contributions: die swell (β_{swell} from m), piston slip (f_{slip} , default 1.0), and recovery ($\sqrt{R_{\text{rec}}}$).
- Recovery is typically the dominant term for hydrogel inks. Low R_{rec} forces $\text{EM} \gg 1$.

ENGINEERING DESIGN RULES

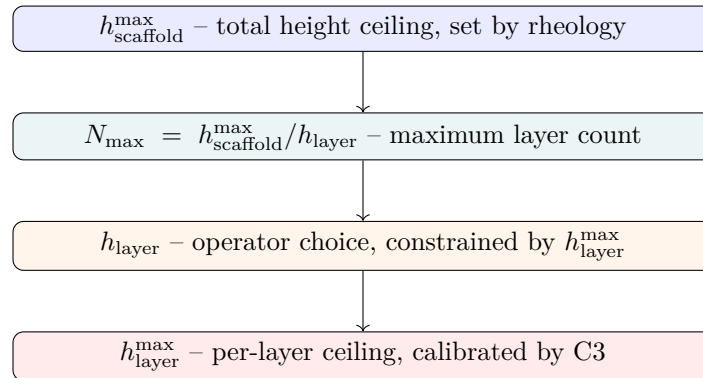
DR 5.1. Always report both k_{flow} and $\text{EM}_{\text{slicer}}$ columns in the lookup CSV so the bench operator sees what to type, not only the diagnostic.

DR 5.2. If $\text{EM} > 1.6$ (e.g. $R_{\text{rec}} < 30\%$), consider whether the ink is suitable for multi-layer prints at all. Very low recovery means significant bead slump and accumulating height errors.

DR 5.3. Measure β_{swell} directly (C1) for any ink with $m < 0.7$. The empirical proxy underestimates swell for strongly viscoelastic inks.

6 The Maximum Scaffold Height

6.1 Three heights, three scales



PHYSICAL INTUITION

$h_{\text{scaffold}}^{\text{max}}$ is about *the whole stack*: can the bottom layer hold everything above it? $h_{\text{layer}}^{\text{max}}$ is about *one layer*: does each fresh layer stay where I put it before the next one arrives? They fail differently:

- Exceeding $h_{\text{scaffold}}^{\text{max}}$: the tower bulges at the base and collapses outward.
- Exceeding $h_{\text{layer}}^{\text{max}}$: each new layer sags into the previous one; the print proceeds but resolution is mushy.

6.2 Deriving $h_{\text{scaffold}}^{\text{max}}$

Per unit plan area, the bottom layer carries the weight of the column:

$$\sigma_{\text{stack}}(N) = \rho g N h_{\text{layer}} = \rho g h_{\text{scaffold}}. \quad (22)$$

For the bottom layer to survive, σ_{stack} must remain below the material's effective strength σ_{max} :

$$h_{\text{scaffold}}^{\text{max}} = \frac{\sigma_{\text{max}}}{\rho g} \quad (23)$$

with $\rho \approx 1000 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$. h_{layer} does *not* appear here: the height ceiling is independent of how thinly you slice.

6.3 What is σ_{max} ? Three criteria

We estimate σ_{max} from SAOS data via three criteria, each appropriate for a different print timescale.

Criterion (A) – LVR endpoint.

$$\sigma_{\text{max}}^{(A)} = G'_{\text{LVR}} \gamma_{\text{LVR}}. \quad (24)$$

Assumes the bottom layer never leaves the linear viscoelastic regime. Appropriate for short-dwell prints where stacking events are brief.

Criterion (B) – $G'(\omega = 1 \text{ rad s}^{-1})$.

$$\sigma_{\text{max}}^{(B)} = G'(\omega = 1 \text{ rad s}^{-1}). \quad (25)$$

The literature default (??). Appropriate for typical multi-layer DIW where dwell time is 1–10 s per layer.

Criterion (C) – $G'(\omega = 0.01 \text{ rad s}^{-1})$.

$$\sigma_{\text{max}}^{(C)} = G'(\omega = 0.01 \text{ rad s}^{-1}), \quad (26)$$

obtained by power-law extrapolation $G'(\omega) = G'_0 \omega^\beta$. Appropriate for slow prints or prints with crosslinking / cell-loading pauses where dwell exceeds $\sim 100 \text{ s}$.

Low- ω gel character. The slope $\beta = d \ln G' / d \ln \omega$ (Winter–Chambon framework ?) diagnoses network permanence:

- $\beta \rightarrow 0$: true gel (permanent network); $B \approx C$.
- $\beta \approx 0.5$: critical gel (transient bonds); $C \ll B$.
- $\beta \rightarrow 1$: viscous liquid (no network).

WORKED EXAMPLE – the three inks

Low- ω fit $G'(\omega) = G'_0 \omega^\beta$:

Ink	G'_0 [Pa]	β [-]	R^2	$G'(\omega = 1)$	$G'(\omega = 0.01)$
C15	114.6	0.454	1.00	114.5	14.1
C15 Gira 5.5	234.9	0.446	0.76	228.4	30.2
Bozano	348.9	0.091	0.98	345.8	229.1

Bozano's $\beta = 0.09$: true gel – retains 66% of its 1 rad/s modulus down to 0.01 rad/s. C15 and NE have $\beta \approx 0.45$: critical-gel regime – they lose modulus on long timescales. Adding NE does not convert C15 to a true gel; it raises the modulus while preserving the network type.

6.4 Three numbers per ink

Selection guide.

- *Short-dwell prints* ($\sim 1 \text{ s/layer}$): criterion (A). NE wins: **56.9 mm**.
- *Typical multi-layer DIW* (1–10 s/layer): criterion (B). Bozano wins: **35.3 mm**.
- *Slow prints* ($> 100 \text{ s/layer}$): criterion (C). Bozano wins decisively: **23.4 mm**.

Ink	(A) LVR endpoint		(B) $G'(\omega = 1)$		(C) $G'(\omega = 0.01)$	
	σ [Pa]	$h_{\max}$ [mm]	σ [Pa]	$h_{\max}$ [mm]	σ [Pa]	$h_{\max}$ [mm]
C15	172.9	17.6	114.5	11.7	14.1	1.4
C15 Gira 5.5	558.5	56.9	228.4	23.3	30.2	3.1
Bozano	47.6	4.8	345.8	35.3	229.1	23.4

Table 1: Maximum scaffold height per ink under three SAOS-derived yield criteria. Bold = the criterion that dominates in the most relevant print regime.

The asymmetry between NE and Bozano. The NE has a wide LVR so its product $G'_{\text{LVR}} \cdot \gamma_{\text{LVR}}$ is large – it wins criterion (A). But its transient network loses modulus at long timescales – it loses (B) and (C). Bozano’s narrow LVR means it loses (A), but its permanent bonds retain modulus – it wins (B) and (C). Neither dominates the other at every timescale: for the manuscript, the NE is the better short-print ink; Bozano is the better long-print ink.

6.5 Lattices and unsupported spans – the Smay criterion

For lattice scaffolds with unsupported strut spans of length L , gravity-driven buckling intervenes earlier than the stacking bound. The Smay criterion (?):

$$h_{\max}(L) = \sqrt{\frac{1.94 G'}{\rho g}} \cdot \sqrt{L}. \quad (27)$$

WORKED EXAMPLE – the three inks

Buckling-corrected $h_{\max}$ at $L = 1$ mm:

Ink	$h_{\max}(L = 1 \text{ mm})$ [mm]
C15	4.76
C15 Gira 5.5	6.72
Bozano	8.27

Even the NE, which can stack to ~ 57 mm as a solid tower, drops to ~ 6.7 mm for a 1-mm lattice span. For lattice scaffolds, the Smay bound is usually the binding constraint.

RUNNING EXAMPLE – Solution to Q5

Question. Will Ana’s $1 \times 1 \times 2 \text{ cm}^3$ block survive its own weight?

From Table 1, C15 Gira 5.5:

- (A) $h_{\max} = 56.9 \text{ mm}$: block at 35% of bound. Easy.
- (B) $h_{\max} = 23.3 \text{ mm}$: block at 86% of bound. Tight.
- (C) $h_{\max} = 3.1 \text{ mm}$: block would collapse at this criterion.

Which criterion applies? At $v_{\text{print}} = 8 \text{ mm/s}$ on a $1 \times 1 \text{ cm}^2$ footprint, layer time $\sim 1\text{--}2 \text{ s}$: criterion **(B) controls**.

Verdict. Feasible but close to the limit. Use $h_{\text{layer}} = 0.36 \text{ mm}$, $N = 56$ layers to reach 20 mm (86% of the (B) bound). Validate in C4.

Still unknown: the Wednesday calibration schedule. Q6.

KEY TAKEAWAYS

- $h_{\text{scaffold}}^{\text{max}} = \sigma_{\text{max}}/(\rho g)$: layer thickness does not appear – slicing into thinner layers does not change the height ceiling.
- The three SAOS criteria span a $10\times$ range; print timescale selects which applies.
- Lattice geometry (S_{may}) is usually the binding constraint for scaffold prints with unsupported spans.

ENGINEERING DESIGN RULES

DR 6.1. Always compute and report all three height criteria. The manuscript should explain which criterion controlled the experiment and why (dwell time argument).

DR 6.2. For lattice designs, compute the S_{may} bound for the intended strut span before committing to an ink. Even a high- G' ink may fail at mm-scale spans.

DR 6.3. Validate $h_{\text{scaffold}}^{\text{max}}$ experimentally in C4 and record which SAOS criterion best matches the observed collapse onset. Use that criterion for all future prints of that ink at similar dwell times.

7 Failure Morphology: Physical Causes and Corrective Actions

This chapter catalogues the failure modes observed in DIW printing, their physical origin in the rheological and hydraulic framework, and the corrective actions available within this workflow. Every corrective action targets V_p , EM_{slicer} , geometry, or needle gauge – never ΔP directly (it is an output, not a knob).

7.1 Under-extrusion

FAILURE MODE: Under-extrusion ($d_{\text{fil}}/\text{ID} < 0.9$)

Observation. Filaments are thinner than the needle ID; gaps appear between adjacent lines; poor adhesion between layers.

Physical causes.

- EM_{slicer} too low: slicer commands too little material per unit length.
- Silent motor stalling: ΔP has exceeded motor stall torque; the piston is not advancing despite E-axis commands.
- Clog or air bubble in the syringe or needle.
- Slip between piston rubber and syringe wall ($f_{\text{slip}} < 1$).

Rheological interpretation. Low R_{rec} gives $k_{\text{flow}} < 1$; if EM_{slicer} was set to k_{flow} instead of $1/k_{\text{flow}}$, the error doubles.

Corrective actions.

1. Raise EM_{slicer} in steps of 0.05.
2. If no change: check predicted ΔP against motor capacity. If close to stall, lower V_p by 25% and switch to wider needle.
3. Purge the needle with slow extrusion; inspect for air.
4. Run C2 to measure f_{slip} directly.

7.2 Over-extrusion

FAILURE MODE: Over-extrusion ($d_{\text{fil}}/\text{ID} > 1.2$)

Observation. Filaments are wider than expected; adjacent lines merge; scaffold walls are rough; towers develop a barrel shape.

Physical causes.

- $\text{EM}_{\text{slicer}}$ too high.
- Die swell larger than predicted by the m -proxy; β_{swell} underestimated.
- V_p too high relative to the XY feedrate (v_{print} too slow).

Corrective actions.

1. Decrease $\text{EM}_{\text{slicer}}$ in steps of 0.05.
2. If EM is already ≤ 1 , reduce V_p by 25%.
3. Measure die swell directly in C1 and update β_{swell} .

7.3 Die swell excess

FAILURE MODE: Filament always wider than needle ID even at low V_p

Physical cause. High stored elastic energy (low m in Cross model or high n in Power Law). The m -based proxy ($\beta \approx 0.3(1 - m)$) may underestimate swell for strongly viscoelastic inks.

Rheological interpretation. High De at exit ($t_{\text{relax}} \gg R_n/\bar{v}_n$): the material exits before it has time to deform viscously; the stored elastic strain releases as radial expansion.

Corrective actions.

1. Measure w_{line} directly in C1 and use $\beta_{\text{swell}}^{\text{exp}} = w_{\text{line}}/\text{ID} - 1$.
2. Use a longer needle (larger L/R): more time at constant flow means more viscous dissipation before exit.
3. Reduce V_p to lower $\bar{v}_n$ and hence the Weissenberg number at exit.

7.4 Scaffold collapse (base bulging)

FAILURE MODE: Base bulging / tower collapse

Observation. The printed object bulges or splays outward from the base as layer count increases.

Physical cause. σ_{stack} has exceeded σ_{max} at the print's effective timescale.

Rheological interpretation. The controlling SAOS criterion has been exceeded. If layer dwell is ~ 1 s, criterion (B) controls; if > 100 s, criterion (C) controls. The NE ink is particularly susceptible on slow prints because $G'(\omega = 0.01) = 30$ Pa predicts $h_{\text{max}} = 3.1$ mm under criterion (C).

Corrective actions.

1. Reduce N to stay under the controlling height criterion.
2. Increase v_{print} to shorten dwell, shifting the controlling criterion from (C) toward (B) or (A).
3. Switch to a higher-modulus ink: Bozano for long-dwell prints; NE for short-dwell.

7.5 Lattice strut buckling

FAILURE MODE: Mid-span sag of bridging filaments

Physical cause. Smay buckling: the unsupported span L is too long for the ink's G' .

Corrective actions.

1. Shorten strut span (reduce cell size).
2. Switch to a higher- G' ink (e.g. Bozano).
3. Increase G'_{LVR} by adding crosslinker or increasing polymer concentration.

7.6 Intermittent extrusion (motor skip)

FAILURE MODE: Intermittent flow / motor skip

Observation. Filament diameter oscillates; audible clicking from the motor; E-axis step count diverges from actual piston position.

Physical cause. Predicted ΔP periodically exceeds motor stall torque, particularly at turns and during acceleration.

Corrective actions.

1. Lower V_p by 25%. Re-check ΔP from CSV.
2. Switch to wider needle (e.g. 21G $\rightarrow$ 18G).
3. Increase acceleration settling time in firmware (add dwell at corners).

7.7 Layer-by-layer fusion (loss of resolution)

FAILURE MODE: Adjacent layers merge; scaffold is mushy

Physical cause. Insufficient recovery between deposition events ($h_{\text{layer}}^{\text{max}}$ exceeded, or R_{rec} too low for the layer dwell).

Corrective actions.

1. Decrease h_{layer} to stay within the C3-validated $h_{\text{layer}}^{\text{max}}$.
2. Increase v_{travel} (non-extrusion move speed) to give the just-deposited layer more time to recover before the next pass.
3. Increase line spacing; reduce infill density.
4. Switch to an ink with higher R_{rec} (Bozano).

7.8 Stringing during travel

FAILURE MODE: Ink trails during non-extrusion moves

Physical cause. Recovery too slow at travel timescale; the ink at the nozzle tip remains mobile during XY travel.

Note. DIW slicers typically do not support retraction (reversing the piston to pull back ink). Attempting retraction with a piston printer causes a pressure deficit that takes many mm of subsequent extrusion to recover – producing a dead zone.

Corrective actions.

1. Ensure retraction is set to **zero** in the slicer.
2. Increase v_{travel} to spend less time dragging the nozzle over the substrate.
3. If stringing persists, raise V_p slightly to increase the viscosity at low $\dot{\gamma}$ (by raising the mean

shear rate in the needle) – counterintuitive but sometimes effective.

7.9 Rough filament surface / extrudate distortion

FAILURE MODE: Rough, wavy, or distorted filament surface

Physical cause. Elastic instability (“die melt fracture” analogue for gels): the Weissenberg number at the needle exit is high enough that the stored elastic energy releases as surface corrugations rather than smooth radial expansion.

Corrective actions.

1. Reduce V_p to lower the Weissenberg number.
2. Use a longer needle to increase viscous dissipation before exit.
3. If the ink is new, check whether η_∞ is very low – a near-zero high-shear plateau makes the exit transition sharp and prone to instability.

KEY TAKEAWAYS

- Every failure mode has a rheological root cause. Diagnose from the parameter tables before adjusting hardware.
- The most common single error is typing k_{flow} into the slicer instead of $1/k_{\text{flow}}$ – produces immediate severe under-extrusion.
- Motor skip at corners is the second most common; always check predicted ΔP against motor capacity before the first print.
- Never use retraction on a piston DIW printer.

ENGINEERING DESIGN RULES

DR 7.1. Diagnose failure morphology *before* adjusting the slicer blindly. Every symptom maps to exactly one physical cause in this chapter.

DR 7.2. Run C1 (filament width sweep) before C2–C4 on any new ink. C1 reveals under/over-extrusion and die swell simultaneously – it is the cheapest diagnostic.

DR 7.3. Keep a photographic log of every calibration session. Failure modes are much faster to diagnose when you can compare current vs. previous prints side-by-side.

Part II – Inks and Instruments

8 The Three Project Inks

8.1 C15 – CMC baseline

Composition. Carboxymethyl cellulose (CMC) in aqueous buffer, formulation **C15**. Molecular weight characterised per ? – *this reference is mandatory in the manuscript; peer review requires it for MW determination.*

Rheological character. Critical-gel polymer solution ($\beta = 0.454$). Pronounced shear thinning ($n_{\text{PL}} = 0.40$, Cross $m = 0.81$). Moderate rest viscosity ($\eta_0 \approx 578$ Pa s). Wide linear regime ($\gamma_{\text{LVR}} = 64\%$). Moderate recovery ($\sim 60\%$ at deposition shear).

Printability profile.

- Pressure: *moderate* ($\Delta P \approx 140$ kPa Cross at $V_p = 0.01$ mm/s, 21G).
- Wall shear: $\tau_w \approx 560$ Pa, $\dot{\gamma}_w \approx 220$ s⁻¹ – cell-safe with margin.
- Scaffold height: modest (12 mm at criterion B).
- Lattice strut bound: tightest of the three (~ 4.8 mm at $L = 1$ mm).
- Extrusion Multiplier: $EM \approx 1.17$.

Use case. Reference / benchmark ink. The simplest formulation; the baseline against which NE inclusion is compared.

8.2 C15 Gira 5.5 – CMC + sunflower-oil nanoemulsion (NE)

Composition. C15 base with statistically optimised sunflower-oil nanoemulsion (project: `Statistical_Analysis_Quantify.tex`) incorporated as a filler.

Rheological character. Critical-gel ($\beta = 0.446$ – microstructure preserved relative to C15). Sharply shear-thinning ($n_{PL} = 0.24$, Cross $m = 0.988$). Quadrupled rest viscosity ($\eta_0 \approx 2440$ Pa s) and tripled LVR modulus ($G'_{LVR} = 916$ Pa) at *unchanged* $\gamma_{LVR} \approx 61\%$. Slightly reduced recovery ($\sim 45\%$).

Printability profile.

- Pressure: modestly higher than C15 ($\Delta P \approx 165$ kPa).
- Wall shear: *lower* than C15 ($\tau_w \approx 660$ Pa, $\dot{\gamma}_w \approx 60$ s⁻¹) – sharper shear thinning.
- Scaffold height: best under criterion A (~ 57 mm); good under criterion B (~ 23 mm).
- Die swell: minimal ($\beta_{swell} \approx 0.004$); cleanest predicted filament geometry.
- Extrusion Multiplier: $EM \approx 1.49$.

Use case. The headline ink of this project. The NE acts as a *reinforcing filler*: it stiffens the network without narrowing the LVR or changing the network type (preserved β , preserved γ_{LVR}). This is the printability advantage to highlight in the manuscript – the NE stiffens without embrittling.

8.3 Bozano – commercial hair gel

Composition. Commercial Bozano hair gel (Gel de Cabelo Bozano) used as control.

Rheological character. *True gel* ($\beta = 0.091$ – the only true gel of the three). Sharply shear-thinning ($n_{PL} = 0.23$, Cross $m = 0.866$). Very high $\eta_0 \approx 6350$ Pa s with the slowest transition ($\lambda = 73$ s). Narrow LVR ($\gamma_{LVR} = 14\%$). Near-complete recovery ($\sim 98\%$ across all tested $\dot{\gamma}$).

Printability profile.

- Pressure: *lowest* of the three ($\Delta P \approx 116$ kPa) – long λ puts the needle deep into the shear-thinned regime.
- Wall shear: lowest ($\tau_w \approx 470$ Pa, $\dot{\gamma}_w \approx 8$ s⁻¹) – extremely benign to cells.
- Scaffold height: loses on criterion A (4.8 mm, narrow LVR penalty) but wins on B (35 mm) and C (23 mm).
- Extrusion Multiplier: $EM \approx 0.93$ – the only ink requiring a *reduction* from nominal.

Use case. Commercial benchmark. Sets the upper bar for recovery and quasi-static stability; sets the lower bar for strain tolerance (narrow LVR). Useful as a reference because it is routinely printed on the lab DIW.

Comparative summary.

Property	C15	C15 Gira 5.5	Bozano
Network type	Critical gel	Critical gel	True gel
η_0 [Pa s]	578	2440	6354
m (sharpness)	0.81	0.99	0.87
G'_{LVR} [Pa]	270	916	348
γ_{LVR} [%]	64	61	14
R_{rec} @ 100 s ⁻¹ [%]	60	45	98
h_{max} criterion B [mm]	12	23	35
EM _{slicer}	1.17	1.49	0.93

9 Nanoemulsion Mechanics

This chapter explains the physical mechanisms by which the sunflower-oil nanoemulsion modifies the CMC matrix. Understanding these mechanisms is essential for interpreting the rheological results, designing improved formulations, and framing the manuscript discussion correctly.

9.1 Nanoemulsion structure and droplet characteristics

The sunflower-oil nanoemulsion consists of oil droplets stabilised by a surfactant shell dispersed in an aqueous continuous phase. Key structural parameters:

- **Droplet size** (d , nm): set by the homogenisation protocol and surfactant HLB value. Smaller droplets ($d < 200$ nm) are kinetically stable and increase the effective particle volume fraction for a given oil loading.
- **Volume fraction** (ϕ): controls the magnitude of viscosity enhancement. At high ϕ , droplets form a jammed network even without polymer bridging.
- **Zeta potential**: determines electrostatic stability. Large negative ζ (< -30 mV) prevents droplet coalescence.

9.2 How the NE stiffens the CMC matrix

Three parallel mechanisms operate when the NE is incorporated into the CMC gel:

Hydrodynamic reinforcement. Rigid or semi-rigid droplets (relative to the surrounding polymer) increase the effective viscosity even without direct polymer-droplet bonds, following the Einstein relation at dilute loading:

$$\eta_{\text{eff}} \approx \eta_{\text{matrix}}(1 + 2.5\phi). \quad (28)$$

At higher ϕ , the Krieger-Dougherty expression replaces this with a divergence near the jamming fraction ϕ_m .

Polymer-droplet bridging. CMC chains adsorb onto the surfactant-coated oil surface. Each droplet effectively becomes a transient crosslink node in the polymer network, increasing G' without forming permanent covalent bonds (hence the preserved γ_{LVR} : a crosslinker would narrow the LVR, but bridging via adsorption does not).

Lubrication at the macroscale. During shear, the oil phase acts as a lubricant between CMC chain bundles, lowering the critical shear rate at which the macroscopic network yields. This is mechanistically responsible for the sharper thinning exponent ($m = 0.988$ for the NE vs. $m = 0.811$ for C15): the oil-mediated lubrication makes the network-to-fluid transition more cooperative.

9.3 Emulsion stability under shear

During extrusion through the needle, the NE is exposed to $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$ for $L_n/\bar{v}_n \approx 0.05 \text{ s}$. Three potential instabilities must be assessed:

- **Droplet deformation:** the capillary number $\text{Ca}_d = \eta_{\text{matrix}} \dot{\gamma} d / \sigma_{\text{NE}}$ must be $< \text{Ca}_d^{\text{crit}} \approx 0.5$ for droplets to remain spherical. For $\eta_{\text{matrix}} \sim 10 \text{ Pa s}$ (at $\dot{\gamma}_w = 60 \text{ s}^{-1}$), $d = 200 \text{ nm}$, $\sigma_{\text{NE}} \sim 5 \text{ mN/m}$: $\text{Ca}_d \sim 0.02 \ll 0.5$. Droplets remain spherical.
- **Droplet coalescence:** time in the high-shear zone is short ($\sim 50 \text{ ms}$). Zeta potential $< -30 \text{ mV}$ provides electrostatic protection. Coalescence risk: low.
- **Ostwald ripening:** relevant only over hours to days in quiescent storage, not during printing. Mitigated by small droplet size and surfactant loading.

Conclusion: the NE is mechanically stable during extrusion at the operating conditions of this project. No special pre-treatment or post-deposition stabilisation is needed for acellular prints.

9.4 Structural recovery enhancement mechanism

The NE reduces R_{rec} relative to C15 (45% vs. 60% at 100 s^{-1}). The mechanism is: after high-shear disruption, the CMC chains must re-entangle and rebuild their network. Droplet re-adsorption is slower than polymer re-entanglement, creating a two-timescale recovery: a fast polymer phase (caught at 30 s) and a slower droplet-bridging phase (minutes). The 30-s measurement used for R_{rec} captures only the first phase. This is why the bead spreads more than expected from the G'_{LVR} alone – the slow recovery phase means the deposited bead is not as stiff as its LVR modulus implies.

Manuscript implication. This two-timescale argument explains why the NE requires a higher $\text{EM}_{\text{ slicer}}$ despite its higher G'_{LVR} . Both must be reported and the mechanism discussed.

9.5 Carrier model rationale

The NE serves as a drug carrier model for future pharmaceutical loading (e.g. Amphotericin B). The oil core provides a hydrophobic domain that sequesters lipophilic drugs; the surfactant shell controls release kinetics and biocompatibility. For the current phase (acellular scaffold printing), no drug is loaded – the NE acts solely as a rheological modifier and structural filler. Future work will characterise the effect of drug loading on the rheological parameters described in this document.

ENGINEERING DESIGN RULES

DR 9.1. Always characterise both the emulsion alone and the CMC+NE composite separately. Changes in droplet size upon mixing must be verified by DLS.

DR 9.2. The preserved γ_{LVR} is the key evidence that the NE acts as a filler rather than a crosslinker. This interpretation must be defended explicitly in the manuscript against reviewers who may suggest chemical crosslinking.

DR 9.3. When loading drugs into the NE, re-characterise η_0 , G'_{LVR} , and R_{rec} – drug molecules can alter the polymer-droplet interfacial energy and change all three.

10 Rheological Tests: Instrument and Battery

10.1 Instrument

Anton Paar MCR 501 stress-controlled rheometer. Plate–plate (PP25, PP50) or cone–plate (CP50) tooling. Solvent trap to suppress evaporation. Temperature 25°C unless noted. Data exported as `.xls` (primary; `xlrd`-compatible – do not use `openpyxl`), `.txt` (human-readable), and `.tri` (proprietary archive, do not alter).

Pre-experiment checklist.

- ☐ Verify instrument zero-gap calibration.
- ☐ Check solvent trap seal; fill if needed.
- ☐ Allow sample to equilibrate at 25 °C for ≥ 10 min before first measurement.
- ☐ Load sample slowly (avoid air bubbles); trim cleanly.
- ☐ Run a brief pre-shear at $\dot{\gamma} = 10 \text{ s}^{-1}$ for 30 s then wait 5 min to erase loading history.

Post-experiment checklist.

- ☐ Export in all three formats; verify file sizes are non-zero.
- ☐ Clean geometry with distilled water immediately; dry completely before storage.
- ☐ Record batch ID, temperature, and gap in the lab notebook.

10.2 Standard test battery

Test	What it measures	Protocol
Flow ramp	$\eta(\dot{\gamma})$, $\tau(\dot{\gamma})$ – constitutive curve	Shear rate ramp $\dot{\gamma} \in [10^{-2}, 10^3] \text{ s}^{-1}$, equilibrium per point.
Flow Sheared Amplitude sweep	Same, after pre-shear G'_{LVR} , γ_{LVR}	200 s^{-1} for 300 s pre-shear, then same ramp. Fixed $\omega = 10 \text{ rad/s}$; sweep γ_0 from 0.01% to $\sim 1000\%$.
Frequency sweep	$G'(\omega)$, $G''(\omega)$	Fixed γ_0 inside LVR (1%); sweep $\omega \in [0.1, 240] \text{ rad/s}$.
Recovery	R_{rec} at deposition $\dot{\gamma}$	Three-interval: $\dot{\gamma}_1 = 0.1$, $\dot{\gamma}_2 \in \{50, 100, 150\} \text{ s}^{-1}$, back to $\dot{\gamma}_1$. Record η at $t = 30 \text{ s}$.

Table 2: Standard test battery. Order: amplitude sweep $\rightarrow$ frequency sweep $\rightarrow$ flow ramp $\rightarrow$ flow sheared $\rightarrow$ recovery. The SAOS tests must be run on a fresh loading (before flow tests destroy the network irreversibly).

Why SAOS before flow tests. Flow ramp and recovery tests apply large strains that partially destroy the network structure. SAOS tests must always be run first on a fresh sample.

10.3 Interpreting pre-shear effects

Compare fitted Cross or Power-Law parameters from Flow vs. Flow Sheared:

- $K_{\text{sheared}} < K$ or $\eta_{0,\text{sheared}} < \eta_0$: pre-shear broke structure that has not fully recovered during the ramp $\Rightarrow$ **thixotropic**.
- $K_{\text{sheared}} > K$: pre-shear organised the network $\Rightarrow$ **rheopexic**.

Both signatures are mechanistically significant and must be discussed in manuscripts. Printability predictions use the *Flow* (no pre-shear) parameters as the default.

10.4 Interpreting the SAOS output: a field guide

- $G' > G''$ at all measured ω : gel-like across all print timescales. Good.
- G'' crosses above G' at low ω (gelation frequency crossover): the material becomes liquid-like at long timescales. Criterion (C) height limit will be small.
- G' flat with ω (low β): true gel. B and C criteria give similar h_{max} .
- G' drops steeply at low ω (high $\beta \approx 0.5$): critical gel / transient network. C criterion is far below B. Avoid slow prints.

11 Extended Material Classes

The framework in Part I was developed for CMC/NE inks, but the equations and protocols apply to any viscoplastic or viscoelastic bioink. This chapter describes how the key parameters change for alternative material classes and what to expect when the framework is applied to them.

11.1 Cellulose-based inks

Hydroxypropyl methylcellulose (HPMC). Similar to CMC in network type (critical gel) but shows *thermogelation*: it thickens on heating rather than cooling. Relevance: printability parameters are temperature-dependent; the Anton Paar temperature program must be validated before fitting.

Bacterial cellulose (BC) nanofibre suspensions. High aspect ratio fibres form entangled networks with very high G'_{LVR} ($\sim 10^3$ – 10^4 Pa) even at low solid content. Die swell is typically small (rigid fibres do not store elastic energy like polymer chains). R_{rec} is high if the network is not covalently crosslinked.

Carboxymethyl cellulose + crosslinker. Adding ionic crosslinkers (e.g. Ca^{2+} for alginate, or citric acid for CMC) *narrows* the LVR and reduces γ_{LVR} , which lowers the LVR-criterion height (criterion A). The true-gel character ($\beta \rightarrow 0$) improves criteria B and C. Crosslinked systems tend to be less tolerant to die swell.

11.2 Alginate-based inks

Alginate crosslinked with Ca^{2+} behaves as a true gel ($\beta < 0.1$). Key differences from CMC/NE:

- η_0 is much higher at comparable solid content; the printable V_p range shifts to lower values.
- R_{rec} is very high ($> 90\%$) post-crosslinking; $\text{EM}_{\text{ slicer}}$ is typically < 1 .
- The Power-Law index n is low (< 0.2); pressure calculations are sensitive to n errors.

11.3 Gelatin-based inks

Gelatin undergoes thermally reversible gelation (melting at $\sim 35^\circ\text{C}$). Rheological testing must be performed at exactly the print temperature (typically 25°C for gelatin-based inks). The Flow ramp must be run with thermal control. The MATLAB simulation is valid without modification, but parameters must be re-entered for each temperature.

11.4 Composite systems: natural + synthetic

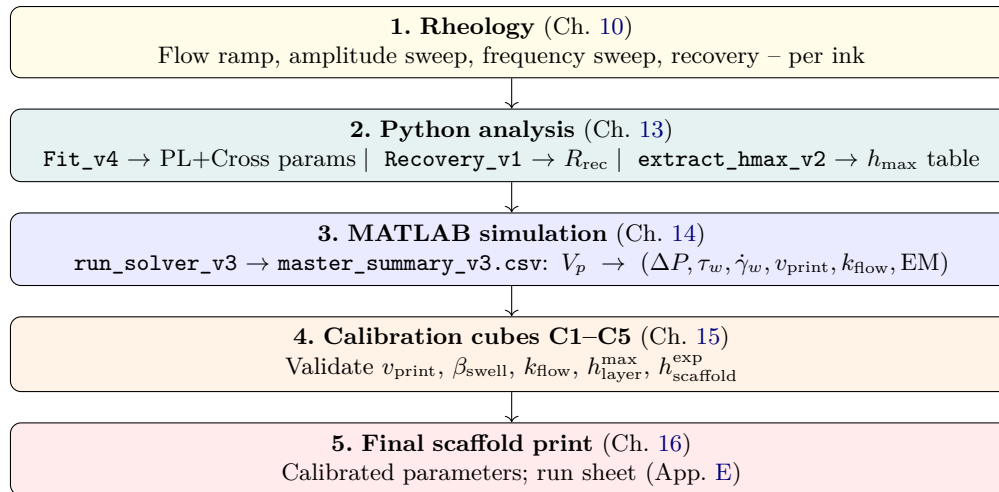
Mixing CMC or alginate with polyethylene glycol (PEG) or gelatin methacrylate (GelMA) introduces:

- A second timescale in $G'(\omega)$ (two-network relaxation).
- Potentially bimodal recovery curves.
- Sensitivity of η_0 to UV crosslinking history.

The framework equations are still valid, but parameter fitting may require a two-mode Cross or Maxwell model. Flag any anomalous $R^2 < 0.95$ on the single-mode Cross fit as a signal of multi-mode behaviour.

Part III – Protocols

12 Workflow Overview



The next sections describe steps 2–5 in operational detail. For step 1, see Ch. 10; for the theoretical rationale of each output, see Part I.

13 Step 2: Python Analysis

13.1 Pre-run checklist

- ☐ Activate the project Python environment.
- ☐ Verify `xlrd`, `scipy`, `numpy`, `pandas`, and `matplotlib` are installed.
- ☐ Check that the `.xls` files from the Anton Paar are in the expected folders (Reologia/Viscosity/, Reologia/Strain/, Reologia/Frequency/, Reologia/Recovery/).
- ☐ Confirm the active script is `Fit_Muitos_Modelos_v4.py` (see CLAUDE.md §6.9).

13.2 Flow-curve fitting – `Fit_Muitos_Modelos_v4.py`

```

cd Analises/Python
python3 Fit_Muitos_Modelos_v4.py
# Outputs: Results/FitAll-Ramp1-v4.txt
# Results/FitAll-Ramp2-v4.txt (if Flow Sheared run)
# Contains: PL (K, n) and Cross (eta_0, eta_inf, lambda, m)
# with R^2, AIC, BIC per model per ink.
  
```

What to check.

- $R^2 \geq 0.98$ for Cross across the full $\dot{\gamma}$ range.
- Cross η_{∞} is finite and positive (not pinned to its lower bound – check that AIC/BIC improved over a 3-parameter fit).
- Cross AIC/BIC lower than PL on log-spaced data.
- PL parameters agree with the Cross slope in the printable $\dot{\gamma}_w$ range (50–1000 s^{−1}).
- If $R^2 < 0.95$ for Cross: suspect multi-mode behaviour or a data quality issue (bubble, edge effects); flag before proceeding.

13.3 Recovery – Recovery_v1.py

```
python3 Recovery_v1.py
# Output: Results/Relatorio_Recovery.txt
# Reports R_rec(%) at t=30s for each (ink, gamma_dot_2).
```

What to check. R_{rec} should decrease monotonically (or remain flat) with increasing $\dot{\gamma}_2$. A non-monotonic pattern (e.g. higher recovery at 100 than at 50 s^{-1}) typically means the recovery test was contaminated by the preceding amplitude sweep (network not fully rebuilt). Repeat with a fresh sample; wait ≥ 5 min between tests.

13.4 Scaffold-height extraction – extract_hmax_v2.py

```
python3 extract_hmax_v2.py
# Outputs:
# Results/SAOS_hmax_v2.txt
# Results/printing_parameters_per_ink.csv
# Results/Gprime_extrap_<ink>.png
# Reports h_scaffold_max under criteria A, B, C and N_max.
```

What to check. The $G'(\omega)$ log-log plot (Gprime_extrap_<ink>.png) should be visually linear at the low- ω end. If the lowest-frequency points scatter, trim the fit window upward by one point and re-run; the low- ω extrapolation is sensitive to noise.

14 Step 3: MATLAB Simulation – run_solver_v3.m

```
cd Analises/MatLab
matlab -batch "run_solver_v3"
# Loops: 3 inks x 2 needles (21G, 22G) x 8 piston velocities.
# Outputs:
# output_v3/slicer_lookup_<ink>_<needle>.csv
# output_v3/plots_<ink>_<needle>.png (4-panel summary)
# output_v3/master_summary_v3.csv (reference row  $V_p=0.01$ )
```

Reading master_summary_v3.csv. Columns (in order): sample, needle, V_p | v_{print} (PL and Cross) | ΔP | τ_w | $\dot{\gamma}_w$ | w_{line} | h_{layer} | k_{flow} | $\text{EM}_{\text{slicer}}$.

SAFETY / DATA INTEGRITY

The CSV column is k_{flow} , *not* $\text{EM}_{\text{slicer}}$. Read the $\text{EM}_{\text{slicer}}$ column (the last column in the lookup CSV, added in v3) for the number to type into the slicer. If you use an older CSV, compute $\text{EM} = 1/k_{\text{flow}}$ by hand. Never type k_{flow} directly into the slicer.

TIP

The default reference row is at $V_p = 0.01 \text{ mm/s}$. For a different operating point, open the per-ink CSV `slicer_lookup_<ink>_<needle>.csv` which covers the full $V_p \in \{0.003 \dots 0.04\} \text{ mm/s}$ sweep.

When to re-run the MATLAB simulation. Any time Python produces new Cross or Power-Law parameters, update the `samples` struct in `run_solver_v3.m` and re-run. Parameters must always be in sync between Python output and MATLAB input.

15 Step 4: Calibration-Cube Protocol (C1–C5)

Run all tests on the same day, with the same syringe load. Each test is a diagnostic for one physical question; if it fails, the failure mode points to one rheological input.

Pre-calibration checklist.

- ☐ Power up printer, home all axes, level bed.
- ☐ Pull ink from 4 °C storage; let it equilibrate at 25 °C for ≥ 30 min.
- ☐ Re-homogenise by slow inversion (30 times); inspect for phase separation or air bubbles.
- ☐ Load syringe; mount needle; prime until steady bead exits.
- ☐ Enter V_p , EM_{slicer} , and h_{layer} from the lookup CSV into the slicer.
- ☐ Verify predicted $\Delta P < 1500$ kPa from CSV.

15.1 C1 – print-speed sweep

C1: identify the optimum v_{print} at fixed V_p

Print five 30 mm straight filaments on a clean glass slide at five XY feedrates $v_{\text{print}} \in \{5, 10, 20, 30, 50\}$ mm/s, keeping V_p fixed at the lookup-CSV value. Wait 5 min. Under a microscope, measure filament width d_{fil} at three points per filament.

Optimum: the v_{print} at which $0.95 \leq d_{\text{fil}}/\text{ID} \leq 1.10$. Below 0.95: head outruns ink supply (under-extrusion or motor skip). Above 1.10: over-deposition or accumulated die swell.

Also read die swell: measure w_{line} at the optimum v_{print} ; compute $\beta_{\text{swell}}^{\text{exp}} = w_{\text{line}}/\text{ID} - 1$ and compare against the proxy from the lookup CSV. Update if $|\beta^{\text{exp}} - \beta^{\text{pred}}| > 0.05$.

15.2 C2 – mass-per-line (k_{flow}) calibration

C2: measure $k_{\text{flow}}^{\text{exp}}$ and update EM

At the C1-optimum v_{print} , print ten parallel 30 mm lines on a tared glass slide with 1 mm spacing. Time the print. Reweigh; convert to volume using $\rho_{\text{ink}} = 1.00$ g/cm³.

$$k_{\text{flow}}^{\text{exp}} = \frac{V_{\text{measured}}}{v_{\text{print}} \cdot w_{\text{line}} \cdot h_{\text{layer}} \cdot t_{\text{print}}}.$$

Invert and enter $EM^{\text{exp}} = 1/k_{\text{flow}}^{\text{exp}}$ as the slicer’s Extrusion Multiplier for all subsequent prints with this ink and needle.

15.3 C3 – layer-height sweep

C3: find $h_{\text{layer}}^{\text{max}}$

Print a 3-layer 10×10 mm square at three layer heights $h_{\text{layer}} \in \{0.30, 0.40, 0.50\}$ mm. After 30 min of relaxation, measure the total stack height.

$h_{\text{layer}}^{\text{max}}$: the largest h_{layer} at which measured total height $\geq 90\%$ of commanded total $3h_{\text{layer}}$.

15.4 C4 – vertical-tower test

C4: validate $h_{\text{scaffold}}^{\text{max}}$ from SAOS

Print a 5×5 mm tower at $h_{\text{layer}}^{\text{max}}$ from C3, up to $N = 40$ layers (or the predicted N_{max} from `printing_parameters_per_ink.csv`, whichever is smaller). Photograph every 5 layers and at $t = 30$ min.

$h_{\text{scaffold}}^{\text{exp}}$: height at which base width grows by $> 10\%$. Record which SAOS criterion best matches the collapse onset. **Use that criterion for all future prints of this ink at similar dwell times.**

15.5 C5 (optional) – bridging test for lattice scaffolds

C5: unsupported-span sag test

Print three pairs of 20 mm pillars at gap lengths $L \in \{0.5, 1.0, 2.0\}$ mm. Print a single bridging filament over each gap; measure mid-span sag after 5 min. Compare against S_{may} prediction (Eq. 27). If sag exceeds the prediction, the strut span must be shortened.

RUNNING EXAMPLE – Solution to Q6

Question. What does Ana’s Wednesday morning look like?

Schedule ($\sim 4\text{--}5$ h):

08:30	Power up; home; level. Pull C15 Gira 5.5 from 4°C ; equilibrate 30 min.
09:00	Re-homogenise; load syringe; prime. Enter $V_p = 0.01$ mm/s, $\text{EM} = 1.49$, $h_{\text{layer}} = 0.36$ mm in slicer. Predicted $\Delta P = 164$ kPa: proceed.
09:30	C1. Five filaments at $v_{\text{print}} \in \{5, 10, 20, 30, 50\}$ mm/s. Microscope. Expected optimum near 10 mm/s.
10:30	C2. Ten lines on tared slide; weigh. Expected $k_{\text{flow}}^{\text{exp}} \in [0.6, 0.8]$; $\text{EM}^{\text{exp}} \in [1.25, 1.67]$.
11:30	C3. Three 3-layer squares at $h_{\text{layer}} \in \{0.30, 0.40, 0.50\}$ mm. Wait 30 min. Expected $h_{\text{layer}}^{\text{max}} \approx 0.40$ mm.
13:00	C4. Tower at $h_{\text{layer}}^{\text{max}}$; target $N = 56$ ($= 20$ mm). Stop early if base bulges.
15:00	Done. Log all calibrated parameters to <code>lab_calibration_log.csv</code> .

Go / No-go. If calibrated parameters match predictions within 30% and C4 tower clears ~ 50 layers cleanly: proceed to Thursday production print. Otherwise: escalate before burning more ink.

16 Step 5: Final Scaffold Print

After C1–C4, the slicer profile contains all five validated entries:

- V_p (E-axis feedrate) at the validated value.
- v_{print} (XY feedrate) at the C1 optimum.
- $\text{EM}^{\text{exp}} = 1/k_{\text{flow}}^{\text{exp}}$ from C2.
- $h_{\text{layer}} \leq h_{\text{layer}}^{\text{max}}$ from C3.
- Total scaffold height $\leq h_{\text{scaffold}}^{\text{exp}}$ from C4.

Save slicer profile as `<ink>_<needle>_<scaffold>_v<n>.ini`. Record the final print on the run sheet (App. E).

17 Troubleshooting: Failure-Mode Decision Tree

18 The Friday Print – Closing the Loop

By Wednesday 15:00, Ana has answered every question this book posed:

- **Q1.** The ink is a reinforced critical gel with wide LVR – excellent DIW candidate.

Observation	Physical root cause	Action
Motor hum / skip; no flow	ΔP exceeds motor stall, or clog	Lower V_p 25%. If still no flow: switch to wider needle; purge.
$d_{\text{fil}}/\text{ID} > 1.2$ consistently	Over-extrusion or β_{swell} underestimated	Lower EM in 0.05 steps; measure β^{exp} in C1.
$d_{\text{fil}}/\text{ID} < 0.9$ consistently	Under-extrusion or motor skip	Raise EM in 0.05 steps; verify ΔP vs. motor limit.
Tower base bulges at low N	σ_{stack} exceeded at effective timescale	Reduce target height; increase v_{print} ; switch ink.
Tower buckles sideways	Smay lattice buckling	Shorten span or use higher- G' ink.
Layer fusion / mushy scaffold	$h_{\text{layer}} > h_{\text{layer}}^{\text{max}}$ or low R_{rec}	Decrease h_{layer} ; increase v_{travel} .
Stringing on travel moves	Slow recovery at travel timescale	Set retraction to zero ; increase v_{travel} .
Rough filament surface	Elastic extrudate instability	Lower V_p ; use longer needle.
First-layer adhesion failure	Bed clean / Z-zero / temperature	Clean bed; raise first-layer h by 0.05 mm; re-home.

Table 3: Operator troubleshooting decision tree. All corrective actions target V_p , EM, geometry, or needle gauge. ΔP is an output and is never adjusted directly.

- **Q2.** $V_p = 0.01$ mm/s to reach $v_{\text{print}} \approx 8$ mm/s.
- **Q3.** $\Delta P \approx 164$ kPa – safe. $\tau_w = 663$ Pa – cell-safe.
- **Q4.** $k_{\text{flow}} \approx 0.67$; EM = 1.49 into the slicer.
- **Q5.** 20-mm block feasible under criterion B; validated by C4.
- **Q6.** Calibration completed 08:30–15:00; no surprises.

Thursday, 10:00. Ana exports G-code, mounts a fresh syringe of conditioned ink, and prints the $1 \times 1 \times 2 \text{ cm}^3$ block. Total print time ≈ 45 min. The block holds its shape. She photographs at $t = 0$, $t = 30$ min, and $t = 24$ h.

Friday, 15:30. Ana hands the supervisor the scaffold and the completed run sheet.

The pedagogical point. Every entry on Ana’s run sheet was computed from a rheological measurement and one equation chain. The slicer firmware was treated as a dumb FFF tool that needed translation. The rheometer told her what the ink wanted; the math told her what to type; the calibration cubes told her where the math was off and by how much. *This is the framework. The rest is execution.*

Part IV – Extended Topics

19 Biological and Regulatory Considerations

19.1 Cell viability and wall shear stress

The primary mechanical threat to cells during DIW extrusion is the wall shear stress τ_w . Literature consensus (?) places the acute viability threshold at:

$$\tau_w^{\text{safe}} \lesssim 5 \text{ kPa}. \quad (29)$$

Above this threshold, cell membrane rupture and cytoskeletal damage increase sharply. For the inks in this project, τ_w at the nominal V_p is 460–660 Pa – one order of magnitude below the threshold. This provides substantial margin for cell-loaded future prints.

Cumulative shear exposure. A single extrusion event (~ 50 ms in the needle) at $\tau_w = 660$ Pa produces a stress dose of $D = \tau_w \cdot t_{\text{needle}} \approx 33$ Pa s. For a 5×5 mm tower with $N = 56$ layers, each cell passes through the needle once. Cumulative dose is therefore 33 Pa s – low. For complex multi-pass prints where cells recirculate through the needle, cumulative dose accumulates and should be estimated before printing.

Cell type sensitivity. Different cell lines have different shear thresholds:

- Osteoblasts and fibroblasts: relatively resistant; tolerate $\tau_w \leq 10$ kPa for brief exposures.
- Stem cells (MSC, iPSC): more sensitive; threshold closer to 5 kPa.
- Chondrocytes: sensitive to both shear and compression history.

For any cell type, measure post-print viability (live/dead assay within 2 h of printing) and compare against unprinted controls. Do not assume compliance from literature thresholds alone.

19.2 Plug flow and cell protection

As noted in Ch. 3, high-Bi inks approach plug flow, with shear concentrated in thin wall layers. This means the majority of cells (in the plug core) experience very low shear stress $\tau \ll \tau_w$. The fraction of cells in the high-shear wall zone can be estimated as:

$$f_{\text{wall}} \approx 1 - \left(\frac{r_{\text{plug}}}{R_n} \right)^2 = 1 - \left(\frac{\tau_y}{\tau_w} \right)^2$$

(for a Bingham fluid). For our inks, even without a formal yield stress, the high η_0 produces a near-plug core that protects most cells.

19.3 Long-duration print considerations

For prints lasting > 30 min with cell-loaded inks:

- Keep the unused ink at 4°C or on ice; avoid temperature fluctuations that change rheology mid-print.
- Monitor filament width at intervals (every 10 layers); drift indicates ink dewatering or temperature change.
- Consider enclosing the printer in a humidified chamber to prevent evaporation from the deposited scaffold.

19.4 MAPA-oriented regulatory considerations

For healthcare applications in Brazil (pharmaceutical or dermocosmetic products), the National Health Surveillance Agency (ANVISA) and the Ministry of Agriculture, Livestock, and Food Supply (MAPA) impose requirements on the production process:

- **Raw material traceability.** The CMC grade, molecular weight determination method (cite ?), and NE formulation parameters (oil type, surfactant, batch number) must all be documented.
- **Process reproducibility.** Calibration data (C1–C4 outputs) must be recorded and archived per batch.

- **Biocompatibility testing.** For dermocosmetic applications (e.g. Amphotericin B carrier), ISO 10993-5 cytotoxicity testing is required. The current project phase is pre-regulatory (acellular scaffold characterisation only).
- **Scale-up validation.** The DIW parameters characterised on the lab-scale piston printer must be re-validated at pilot scale; scale-up introduces pressure gradients and syringe-wall effects not present at 10 mL scale.

TIP

When writing the manuscript, frame the DIW as a “proof-of-concept fabrication platform” rather than a production process. This framing keeps the paper in the research domain and defers the regulatory validation discussion to future work.

20 Statistical Optimisation: DOE and Response Surface Methodology

20.1 Overview

Design of Experiments (DOE) and Response Surface Methodology (RSM) are the standard tools for optimising the NE formulation (oil fraction, surfactant type, surfactant concentration, homogenisation protocol) with respect to measured responses (droplet size, η_0 , G'_{LVR} , printability). This section documents the critical decisions required before running any RSM, with emphasis on categorical variables – the most common source of invalid models in this type of study.

20.2 Continuous vs. categorical variables

RSM models assume all factors are *continuous*: they can take any value in a range, and the response varies smoothly with them. A factor is *categorical* if it takes discrete, unordered levels with no implied metric distance between them.

Continuous factors (safe for RSM quadratic terms):

- Oil mass fraction [%]
- Surfactant concentration [mg/mL]
- Homogenisation speed [rpm]
- Homogenisation time [min]

Categorical factors (cannot receive quadratic terms in RSM):

- Surfactant type (Tween 80 / Span 80 / Polysorbate 20 / ...)
- Emulsification method (probe sonication / high-pressure HPH / rotor-stator)
- Needle type (blunt / tapered)
- Pulse mode (on/off), if coded as 1/0

SAFETY / DATA INTEGRITY

Categorical variables must not appear in quadratic RSM terms. A quadratic term for “surfactant type” (e.g. $x_{\text{surfactant}}^2$ where surfactant type is coded as 1, 2, 3) imposes an implied quadratic distance between surfactant types that has no physical meaning. The resulting “optimal” surfactant is often a non-existent interpolation between two real options.

Correct handling: Run separate RSM surfaces for each level of the categorical factor, or use mixture designs, or test only one categorical level at a time with a fixed level for all others.

20.3 Model selection recommendations

- For 2–4 continuous factors: full Central Composite Design (CCD) or Box-Behnken Design. Minimum 5 centre-point replicates for pure error estimation.
- For > 4 continuous factors: use a D-optimal or I-optimal custom design to keep run count manageable.
- For one categorical factor: use a split-plot design or run separate CCD blocks per categorical level.
- Always report the lack-of-fit test. A significant lack of fit ($p < 0.05$) means the quadratic surface does not capture the response – add cross terms or increase the polynomial degree.

20.4 Responses and their transformations

- **Droplet size** (d , nm): typically log-normally distributed. Fit after $\log_{10}$ transformation; check residuals.
- **Viscosity** (η_0 , Pa s): strongly right-skewed. Use $\ln(\eta_0)$ as the response.
- **Modulus** (G'_{LVR} , Pa): similarly skewed. Use $\ln(G'_{\text{LVR}})$.
- **Recovery** (R_{rec} , %): bounded $[0, 100]$. If values cluster near the bounds, use logit transformation: $\text{logit}(p) = \ln(p/(1 - p))$.

A Glossary

V_p – piston velocity

Linear advance speed of the syringe piston in mm/s. The single flow input on this printer.

Q – volumetric flow rate

$$Q = \pi R_s^2 V_p \text{ (mm}^3/\text{s)}.$$

$\bar{v}_n$ – mean nozzle velocity

$$Q/(\pi R_n^2): \text{ average velocity inside the needle (mm/s).}$$

v_{print} – print speed

$$Q/(w_{\text{line}} h_{\text{layer}}): \text{ head XY translation speed during deposition.}$$

ΔP – extrusion pressure

Pressure drop across the needle. **Output** on this printer; diagnostic against the ~ 1500 kPa syringe ceiling.

τ_w – wall shear stress

Force per unit area at the needle wall. Cell-safe below ~ 5 kPa.

$\dot{\gamma}_w$ – wall shear rate

Strain rate at the needle wall; selects the recovery test shear rate.

K, n – Power-Law parameters

$$\tau = K \dot{\gamma}^n.$$

$\eta_0, \eta_\infty, \lambda, m$ – Cross-model parameters

Eq. 6.

G' – storage modulus

Elastic part of SAOS response (Pa).

G'' – loss modulus

Viscous part of SAOS response (Pa).

 $\tan \delta$

G''/G' : loss tangent. < 1 = gel-like.

 γ_{LVR} – LVR endpoint strain

Strain at which G' has dropped 10% from its plateau in an amplitude sweep [%].

 β – low- ω slope

$d \ln G' / d \ln \omega$ at low frequency. Gel-character diagnostic: $\beta \rightarrow 0$ = true gel, $\beta \approx 0.5$ = critical gel.

 R_{rec} – structural recovery

% return of η at 30 s after high-shear disruption.

 h_{layer}

Layer thickness chosen by the operator.

 $h_{\text{layer}}^{\text{max}}$

Largest h_{layer} validated in C3 (per-layer sag bound).

 $h_{\text{scaffold}}^{\text{max}}$

Total height ceiling from $\sigma_{\text{max}}/(\rho g)$.

 N_{max}

$h_{\text{scaffold}}^{\text{max}}/h_{\text{layer}}$.

 k_{flow} – deposition efficiency

Fraction of the slicer-commanded volume that is locked in the bead (Eq. 19). **Not** the number to type into the slicer.

 $\text{EM}_{\text{slicer}}$ – Extrusion Multiplier

$1/k_{\text{flow}}$ (Eq. 20). This is the number entered into the slicer (labelled “Flow”, “Flow Rate”, or “Volumetric Flow Compensation” depending on the slicer package).

 β_{swell} – die-swell ratio

$w_{\text{line}}/\text{ID} - 1$. Estimated from Cross m or measured in C1.

LVR

Linear viscoelastic regime.

SAOS

Small-amplitude oscillatory shear.

Critical gel

Transient network on the edge of percolation ($\beta \approx 0.5$).

True gel

Permanent network ($\beta \rightarrow 0$).

Rabinowitsch–Weissenberg integral

Generalises Hagen–Poiseuille to any $\eta(\dot{\gamma})$. Eq. 16.

Smay criterion

Buckling bound for unsupported lattice struts. Eq. 27.

Die swell

Radial expansion of the extrudate on exit from the needle, driven by elastic stress release.

Deborah number De

$t_{\text{relax}}/t_{\text{process}}$; governs whether elastic or viscous behaviour dominates.

Bingham number Bi

Ratio of yield stress to viscous stress; governs whether flow is plug-like or Newtonian-like.

Capillary number Ca

Ratio of viscous to surface tension forces.

DOE

Design of Experiments.

RSM

Response Surface Methodology.

B Notation and Unit Reference

Symbol	Name	SI unit	Practical unit
V_p	Piston velocity	m/s	mm/s
Q	Volumetric flow rate	m ³ /s	mm ³ /s
$\bar{v}_n$	Mean nozzle velocity	m/s	mm/s
v_{print}	Print speed	m/s	mm/s
R_s	Syringe inner radius	m	mm
R_n	Needle inner radius	m	mm
L_n	Needle length	m	mm
τ	Shear stress	Pa	Pa
τ_w	Wall shear stress	Pa	Pa
$\dot{\gamma}$	Shear rate	s ⁻¹	s ⁻¹
η	Viscosity	Pa s	Pa s
η_0	Zero-shear viscosity	Pa s	Pa s
ΔP	Pressure drop	Pa	kPa
G'	Storage modulus	Pa	Pa
G''	Loss modulus	Pa	Pa
K	Power-Law consistency	Pas ^{n}	Pas ^{n}
n	Power-Law index	–	–
λ	Cross relaxation time	s	s
m	Cross exponent	–	–
ρ	Ink density	kg/m ³	g/cm ³
h_{layer}	Layer thickness	m	mm
$h_{\text{scaffold}}^{\text{max}}$	Max scaffold height	m	mm
k_{flow}	Deposition efficiency	–	–
EM	Extrusion Multiplier	–	–
β_{swell}	Die-swell ratio	–	–
R_{rec}	Structural recovery	–	%

Hardware defaults (override in CLAUDE.md §6.5 if changed):

Component	ID (mm)	Length (mm)
Syringe (BD 10 mL)	14.30 ($R_s = 7.15$)	90
Needle 21G blunt	0.515 ($R_n = 0.258$)	31.75
Needle 22G blunt	0.413 ($R_n = 0.207$)	25.40

C Worked End-to-End Example: C15 Gira 5.5 on 21G**Step 0 – starting from rheology.**

- Cross: $\eta_0 = 2440$, $\eta_{\infty} = 0.001$, $\lambda = 4.10$ s, $m = 0.988$.

- PL: $K = 267$, $n = 0.24$.
- SAOS: $G'_{\text{LVR}} = 916 \text{ Pa}$, $\gamma_{\text{LVR}} = 60.9\%$, $G'(\omega = 1) = 228 \text{ Pa}$, $G'(\omega = 0.01) = 30 \text{ Pa}$.
- Recovery at 100 s^{-1} : $R_{\text{rec}} = 45\%$.

Step 1 – $V_p \rightarrow Q$. $V_p = 0.01 \text{ mm/s}$; $Q = \pi(7.15)^2 \cdot 0.01 = 1.6 \text{ mm}^3/\text{s}$.

Step 2 – Cross-model ΔP . Rabinowitsch–Weissenberg with 21G ($R_n = 0.258 \text{ mm}$, $L_n = 31.75 \text{ mm}$): $\tau_w \approx 660 \text{ Pa}$, $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$, $\Delta P \approx 164 \text{ kPa}$.

Step 3 – safety check. $\Delta P = 164 \text{ kPa} \ll 1500 \text{ kPa}$. $\tau_w = 660 \text{ Pa} \ll 5 \text{ kPa}$. Print is mechanically and biologically feasible.

Step 4 – slicer parameters. $\beta_{\text{swell}} = 0.3(1 - 0.988) = 0.004$; $w_{\text{line}} = 0.52 \text{ mm}$; $h_{\text{layer}} = 0.7 \cdot 0.515 = 0.36 \text{ mm}$; $k_{\text{flow}} = (1.004)^2 \cdot 1 \cdot \sqrt{0.45} = 0.67$; $\text{EM}_{\text{slicer}} = 1/0.67 \approx 1.49$.

Step 5 – $h_{\text{scaffold}}^{\text{max}}$. Criterion (B) at typical 1-s/layer dwell: $h_{\text{max}} = 228/(1000 \cdot 9.81) \cdot 10^3 = 23.3 \text{ mm}$ (65 layers at $h_{\text{layer}} = 0.36 \text{ mm}$).

Step 6 – expected calibration results. C1: $v_{\text{print}} \approx 8\text{--}10 \text{ mm/s}$. C2: $k_{\text{flow}}^{\text{exp}} \approx 0.60\text{--}0.80$; $\text{EM}^{\text{exp}} \approx 1.25\text{--}1.67$. C3: $h_{\text{layer}}^{\text{max}} \approx 0.40 \text{ mm}$. C4: tower to $N \approx 60$ layers; best-matching criterion: (B).

Step 7 – final scaffold. $5 \times 5 \times 20 \text{ mm}$ solid block at 21G is comfortably within all bounds and is the recommended first production print for this ink.

D Engineering Design Rules – Consolidated Reference

- DR 1.** Before reading any DIW paper, identify the control mode (pneumatic, screw, or piston). Parameters are not transferable without going through the constitutive model.
- DR 2.** Never enter ΔP into the slicer – it is an output on this printer.
- DR 3.** Always report both G'_{LVR} and γ_{LVR} ; their product controls scaffold height.
- DR 4.** Prefer the Cross model for printability prediction. Use the Power Law only for single-decade fitting.
- DR 5.** Structural recovery is more important than static viscosity for scaffold quality.
- DR 6.** Measure β_{swell} directly for any ink with $m < 0.7$.
- DR 7.** $\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$. Never type k_{flow} into the slicer.
- DR 8.** Verify $\text{Re}_{\text{gen}} \ll 2100$ before accepting any ΔP calculation.
- DR 9.** Do not change needle gauge without recomputing all downstream quantities; $\Delta P \propto R_n^{-4}$.
- DR 10.** For cell-loaded inks, prefer high-Bi inks (approaching plug flow).
- DR 11.** Always report ΔP , τ_w , and $\dot{\gamma}_w$ as a trio.
- DR 12.** Always compute and report all three h_{max} criteria.
- DR 13.** For lattices, always compute the Smay bound for the intended strut span.
- DR 14.** Validate $h_{\text{scaffold}}^{\text{max}}$ in C4 and record which criterion best matches.

- DR 15.** Diagnose failure morphology from first principles before adjusting the slicer blindly.
- DR 16.** Run C1 before C2–C4 on any new ink.
- DR 17.** Set retraction to zero on any piston DIW printer.
- DR 18.** Update MATLAB parameters immediately when Python produces new Cross/PL parameters.
- DR 19.** Categorical variables must not appear in quadratic RSM terms.
- DR 20.** Preserve the daSilva2018 citation for CMC molecular weight in every manuscript.

E Single-Page Run Sheet

DIW Printing Run Sheet – Textbook v2.0					
Operator:_____		Date:_____		Ink batch:_____	
Ink: ☐ C15 ☐ C15 Gira 5.5 ☐ Bozano		Needle: ☐ 21G ☐ 22G		Room T (°C):_____	
Predicted (from CSV):					
V_p =_____ mm/s		v_{print} =_____ mm/s		ΔP =_____ kPa τ_w =_____ Pa k_{flow} =_____	
EM = $1/k$ =_____					
h_{layer} =_____ mm		$N_{\text{max}}(\text{B})$ =_____		$N_{\text{max}}(\text{C})$ =_____ $\Delta P < 1500$ kPa? ☐ Yes ☐ No	
(stop)					
C1 line widths (d_{fil}/ID) at 5:_____ 10:_____ 20:_____ 30:_____ 50:_____					
Optimum v_{print} :_____ mm/s		$\beta_{\text{swell}}^{\text{exp}}$:_____			
C2: m_{dep} =_____ mg		t_{print} =_____ s		$k_{\text{flow}}^{\text{exp}}$ =_____ EM ^{exp} = $1/k$ =_____	
C3: stack @ 0.30:_____		@ 0.40:_____		@ 0.50:_____ $h_{\text{layer}}^{\text{max}}$ =_____ mm	
C4: N^{exp} =_____		$h_{\text{scaffold}}^{\text{exp}}$ =_____ mm		Best criterion: ☐ A ☐ B ☐ C	
Motor stall events: ☐ None ☐ _____					
Final slicer profile:		V_p =_____ v_p =_____		EM=_____ h =_____ Profile	
file:_____					
Notes:_____					

F Software Setup Guide

F.1 Python environment

```
# Create environment
python3 -m venv .venv
source .venv/bin/activate # macOS/Linux
# .venv\Scripts\activate # Windows

# Install dependencies
pip install numpy scipy pandas xlrd matplotlib

# Verify xlrd version (must be 1.x for .xls; 2.x removed .xls support)
pip show xlrd # should show Version: 1.x.x
# If 2.x: pip install xlrd==1.2.0
```

Key dependency constraint. The Anton Paar .xls files use the legacy binary format. The xlrd library dropped support for this format in version 2.0. Always use xlrd==1.2.0. Never substitute openpyxl – it reads .xlsx only.

F.2 MATLAB

MATLAB R2021a or later is required. The following toolboxes are used:

- Optimization Toolbox (`fzero`)
- Statistics and Machine Learning Toolbox (optional, for AIC/BIC)

No additional file-path configuration is needed; the driver scripts use relative paths from `Analises/MatLab/`.

F.3 LaTeX

This document was compiled with pdfLaTeX (TeX Live 2023+). Required packages: `amsmath`, `amssymb`, `booktabs`, `siunitx`, `tcolorbox`, `tikz`, `natbib`, `lastpage`, `titlesec`, `fancyhdr`, `enumitem`. Install via `tlmgr` or use an Overleaf project with TeX Live 2023 compiler.